

Exercise J-A

1. The integral $\int \frac{\sec^2 x}{(\sec x + \tan x)^{9/2}} dx$ equals (for some arbitrary constant K) [JEE-2012]

समाकलन $\int \frac{\sec^2 x}{(\sec x + \tan x)^{9/2}} dx$ बराबर होगा (किसी स्वेच्छ नियतांक K के लिए) [JEE-2012]

(A) $-\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} - \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$ (B) $\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} - \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$

(C) $-\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} + \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$ (D) $\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} + \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$

Ans. C

Sol. Let $I = \int \frac{\sec^2 x}{(\sec x + \tan x)^{9/2}} dx$

$$= \int \frac{\sec x (\sec x + \tan x) \sec x}{(\sec x + \tan x)^{11/2}} dx$$

Put $\sec x + \tan x = t$

$$\Rightarrow (\sec x \tan x + \sec^2 x) dx = dt$$

Also $\because \sec^2 x - \tan^2 x = 1$

$$\Rightarrow \sec x - \tan x = \frac{1}{t}$$

$$\therefore \sec x = \frac{1}{2} \left(t + \frac{1}{t} \right)$$

$$\therefore I = \frac{1}{2} \int \frac{\left(t + \frac{1}{t} \right) dt}{t^{11/2}} = \frac{1}{2} \int (t^{-9/2} + t^{-13/2}) dt$$

$$= \frac{1}{2} \left(-\frac{2t^{-7/2}}{7} - \frac{2t^{-11/2}}{11} \right) + K$$

$$= -\frac{1}{t^{11/2}} \left(\frac{1}{11} + \frac{t^2}{7} \right) + K$$

$$= -\frac{1}{(\sec x + \tan x)^{11/2}} \left\{ \frac{1}{11} + \frac{1}{7} (\sec x + \tan x)^2 \right\} + K$$

2. The value of the integral $\int_{-\pi/2}^{\pi/2} \left(x^2 + \ln \frac{\pi+x}{\pi-x} \right) \cos x \, dx$ is : [JEE-2012]

$\int_{-\pi/2}^{\pi/2} \left(x^2 + \ln \frac{\pi+x}{\pi-x} \right) \cos x \, dx$ का मान होगा : [JEE-2012]

- (A) 0 (B) $\frac{\pi^2}{2} - 4$ (C) $\frac{\pi^2}{2} + 4$ (D) $\frac{\pi^2}{2}$

Ans. B

Sol.
$$\begin{aligned} & \int_{-\pi/2}^{\pi/2} x^2 \cos x \, dx + \int_{-\pi/2}^{\pi/2} \ln \left(\frac{\pi+x}{\pi-x} \right) \cos x \, dx \\ &= \int_{-\pi/2}^{\pi/2} x^2 \cos x \, dx = 2 \int_0^{\pi/2} x^2 \cos x \, dx \\ &= 2 \left((x^2 \sin x)_0^{\pi/2} - 2 \int_0^{\pi/2} x \sin x \, dx \right) \\ &= 2 \left(\frac{\pi^2}{4} - 2 \left(-(x \cos x)_0^{\pi/2} + \int_0^{\pi/2} \cos x \, dx \right) \right) \\ &= 2 \left(\frac{\pi^2}{4} - 2 \int_0^{\pi/2} \cos x \, dx \right) \\ &= 2 \left(\frac{\pi^2}{4} - 2 \right) = \frac{\pi^2}{2} - 4 \end{aligned}$$

3. For $a \in \mathbb{R}$ (the set of all real numbers), $a \neq -1$, $\lim_{n \rightarrow \infty} \frac{(1^a + 2^a + \dots + n^a)}{(n+1)^{a-1} [(na+1) + (na+2) + \dots + (na+n)]} = \frac{1}{60}$. Then $a =$ [JEE(Adv.)2013]

$a \in \mathbb{R}$ (सभी वास्तविक संख्याओं का समुच्चय), $a \neq -1$, के लिए $\lim_{n \rightarrow \infty} \frac{(1^a + 2^a + \dots + n^a)}{(n+1)^{a-1} [(na+1) + (na+2) + \dots + (na+n)]} = \frac{1}{60}$

तब $a =$

[JEE(Adv.)2013]

- (A) 5 (B) 7 (C) $\frac{-15}{2}$ (D) $\frac{-17}{2}$

Ans. B

Sol.
$$L = \lim_{n \rightarrow \infty} \frac{1^a + 2^a + \dots + n^a}{(n+1)^{a-1} \left[\underbrace{na + na + \dots + na}_{n \text{ times}} + 1 + 2 + 3 + \dots + n \right]} = \lim_{n \rightarrow \infty} \frac{\sum_{r=1}^n r^a}{(n+1)^{a-1} \left[n^2 a + \frac{n(n+1)}{2} \right]}$$

$$= \lim_{n \rightarrow \infty} \frac{\left(\frac{1}{n} \sum_{r=1}^n \frac{r^a}{n^a} \right) n^{a+1}}{(n+1)^{a-1} \left[n^2 a + \frac{n(n+1)}{2} \right]} = \lim_{n \rightarrow \infty} \frac{\left(\frac{1}{n} \sum_{r=1}^n \frac{r^a}{n^a} \right)}{\left(\frac{n+1}{n} \right)^{a-1} \left[\frac{n^2 a + \frac{n(n+1)}{2}}{n^2} \right]}$$

$$= \frac{\int_0^1 x^a dx}{\left(a + \frac{1}{2} \right)} = \frac{1}{60} \Rightarrow \frac{2}{(a+1)(2a+1)} = \frac{1}{60}$$

$$\Rightarrow 2a^2 + 3a - 119 = 0 \Rightarrow a = 7 \text{ \& } -\frac{17}{2}$$

$$a = -\frac{17}{2} \text{ will be rejected as } \int_0^1 x^{-\frac{17}{2}} dx \text{ is not defined.}$$

$$\Rightarrow a = 7$$

4. Let $f: [a, b] \rightarrow [1, \infty)$ be a continuous function and let $g: \mathbb{R} \rightarrow \mathbb{R}$ be defined as
$$g(x) = \begin{cases} 0 & \text{if } x < a, \\ \int_a^x f(t) dt & \text{if } a \leq x \leq b, \\ \int_a^b f(t) dt & \text{if } x > b \end{cases}$$

Then:

[JEE(Adv.)2014]

- (A) $g(x)$ is continuous but not differentiable at a
- (B) $g(x)$ is differentiable on $\mathbb{R}$
- (C) $g(x)$ is continuous but not differentiable at b
- (D) $g(x)$ is continuous and differentiable at either a or b but not both

माना कि $f: [a, b] \rightarrow [1, \infty)$ एक सतत फलन है तथा $g: \mathbb{R} \rightarrow \mathbb{R}$ निम्नानुसार $g(x) = \begin{cases} 0 & \text{if } x < a, \\ \int_a^x f(t) dt & \text{if } a \leq x \leq b, \\ \int_a^b f(t) dt & \text{if } x > b \end{cases}$ परिभाषित है तब:

[JEE(Adv.)2014]

(A) a पर $g(x)$ सतत (continuous) है परन्तु अवकलनीय नहीं है

(B) $\mathbb{R}$ पर $g(x)$ अवकलनीय है

(C) b पर $g(x)$ सतत है परन्तु अवकलनीय नहीं है।

(D) a या b पर $g(x)$ सतत एवं अवकलनीय है परन्तु दोनों पर नहीं

Ans. AC

Sol. $\lim_{x \rightarrow a^+} g(x) = 0$

$$\lim_{x \rightarrow a^+} g(x) = \lim_{x \rightarrow a^+} \int_a^x f(t) dt = \int_a^a f(t) dt = 0$$

$$g(a) = \int_a^a f(t) dt = 0$$

Similarly at $x = b$ we will be get continuous

So $g(x)$ is continuous $\forall x \in \mathbb{R}$

$$g'(x) = \begin{cases} 0 & x < a \\ f(x) & a \leq x \leq b \\ 0 & x > b \end{cases}$$

$$g'(a^-) = 0 \quad g'(b^-) = f(b)$$

$$g'(a^+) = f(a) \quad g'(b^+) = 0$$

Since $f(x)$ co-domain is $[1, \infty)$ $f(a)$ and $f(b)$ can never be zero.

Hence it is non derivable at $x = a$ and $x = b$

5. The value of $\int_0^1 4x^3 \left\{ \frac{d^2}{dx^2} (1-x^2)^5 \right\} dx$ is :

[JEE(Adv.)2014]

निम्न $\int_0^1 4x^3 \left\{ \frac{d^2}{dx^2} (1-x^2)^5 \right\} dx$ का मान है :

[JEE(Adv.)2014]

Ans. 2

Sol. $\left(4x^3 \cdot \frac{d}{dx}(1-x^2)^5\right)_0^1 - 12 \int_0^1 x^2 \cdot \frac{d}{dx}(1-x^2)^5 dx$

$$= \left[4x^3 \cdot 5(1-x^2)^4 \cdot -2x\right]_0^1 - 12 \left[x^2(1-x^2)^5\right]_0^1 + 24 \int_0^1 x(1-x^2)^5 dx = 2$$

Alternative :

$$\int_0^1 4x^3 \left\{ \frac{d^2}{dx^2}(1-x^2)^5 \right\} dx$$

$$\frac{d}{dx} \left(\frac{d(1-x^2)^5}{dx} \right) = \frac{d}{dx} (5(1-x^2)^4 (-2x))$$

$$= -10 \frac{d}{dx} (x(1-x^2)^4)$$

$$= -10 [(1-x^2)^4 + x^4(1-x^2)^3 (-2x)]$$

$$= [-10(1-x^2)^3] [1-x^2-8x^2]$$

Hence Integral

$$= -40 \int_0^1 x^3 (1-x^2)^3 (1-9x^2) dx \quad \text{Put } x = \sin \theta$$

$$= -40 \int_0^{\pi/2} \sin^3 \theta \cos^7 \theta d\theta + 360 \int_0^1 \sin^5 \theta \cos^7 \theta d\theta$$

$$= -40 \cdot 1 \cdot \frac{2 \cdot 6 \cdot 4 \cdot 2}{10 \cdot 8 \cdot 6 \cdot 2} + 360 \cdot \frac{4 \cdot 2 \cdot 6 \cdot 4 \cdot 2}{12 \cdot 10 \cdot 8 \cdot 6 \cdot 4 \cdot 2} = -1 + 3 = 2$$

6. The following integral $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (2 \operatorname{cosec} x)^{17} dx$ is equal to : **[JEE(Adv.)2014]**

निम्न समाकल $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (2 \operatorname{cosec} x)^{17} dx$ नीचे दिये गये विकल्पों में से किसके समान है : **[JEE(Adv.)2014]**

(A) $\int_0^{\log(1+\sqrt{2})} 2(e^u + e^{-u})^{16} du$

(B) $\int_0^{\log(1+\sqrt{2})} (e^u + e^{-u})^{17} du$

(C) $\int_0^{\log(1+\sqrt{2})} (e^u - e^{-u})^{17} du$

(D) $\int_0^{\log(1+\sqrt{2})} 2(e^u - e^{-u})^{16} du$

Ans. A

Sol. $I = \int_{\pi/4}^{\pi/2} (2 \operatorname{cosec} x)^{17} dx$

Put $\ln \tan \frac{x}{2} = t \Rightarrow \tan \frac{x}{2} = e^t$

$$\Rightarrow \sin x = \frac{2e^t}{1+e^{2t}}$$

$$\operatorname{cosec} x = \frac{e^t + e^{-t}}{2}$$

$$I = 2 \int_{\ln(\sqrt{2}-1)}^0 (e^t + e^{-t})^{16} dt$$

$$= 2 \int_{-\ln(\sqrt{2}+1)}^0 (e^t + e^{-t})^{16} dt$$

since $(e^t + e^{-t})^{16}$ is an even function

Hence $I = \int_0^{\ln(\sqrt{2}+1)} 2(e^t + e^{-t})^{16} dt$

7. Let $f: [0,2] \rightarrow \mathbb{R}$ be a function which is continuous on $[0,2]$ and is differentiable on $(0,2)$ with $f(0) = 1$. Let $F(x) = \int_0^{x^2} f(\sqrt{t}) dt$ for $x \in [0,2]$. If $F'(x) = f(x)$ for all $x \in (0,2)$, then $F(2)$ equals : **[JEE(Adv.)2014]**

माना कि $f: [0,2] \rightarrow \mathbb{R}$ एक ऐसा फलन है जो $[0,2]$ पर संतत है एवं $(0,2)$ पर अवकलनीय है तथा $f(0) = 1$ है। माना कि सभी $x \in [0,2]$ के लिये माना $F(x) = \int_0^{x^2} f(\sqrt{t}) dt$ है। यदि सभी $x \in (0,2)$ के लिये $F'(x) = f(x)$ है, तब $F(2)$ का मान है :

[JEE(Adv.)2014]

- (A) $e^2 - 1$ (B) $e^4 - 1$ (C) $e - 1$ (D) e^4

Ans. B

Sol. $F'(x) = f(x) \cdot 2x$

$$\Rightarrow \frac{F'(x)}{f(x)} = 2x$$

$$\Rightarrow \int \frac{f'(x)}{f(x)} dx = \int 2x dx \left\{ \because f'(x) = F'(x) \right\}$$

$$\Rightarrow F(x) = e^{x^2} + C$$

$$\therefore F(0) = 0 \Rightarrow C = -1$$

$$\Rightarrow F(x) = e^{x^2} - 1$$

$$\Rightarrow F(2) = e^4 - 1$$

Paragraph For Question Nos. 8 to 9

Given that for each $a \in (0, 1)$, $\lim_{h \rightarrow 0^+} \int_h^{1-h} t^{-a} (1-t)^{a-1} dt$ exists. Let this limit be $g(a)$. In addition, it is given that the function $g(a)$ is differentiable on $(0, 1)$.

दिया गया है कि प्रत्येक $a \in (0, 1)$ के लिये सीमा $\lim_{h \rightarrow 0^+} \int_h^{1-h} t^{-a} (1-t)^{a-1} dt$ अस्तित्व में है। माना कि यह सीमा $g(a)$ है। इसके अतिरिक्त यह भी दिया गया है कि अन्तराल $(0, 1)$ पर फलन $g(a)$ अवकलनीय है।

8. The value of $g\left(\frac{1}{2}\right)$ is : [JEE(Adv.)2014]

$g\left(\frac{1}{2}\right)$ का मान है : [JEE(Adv.)2014]

(A) π (B) 2π (C) $\pi/2$ (D) $\pi/4$

Ans. A

Sol. Using Leibniz rule

$$g'(a) = \int_h^{1-h} \frac{\delta}{\delta a} t^{-a} (1-t)^{a-1} dt$$

$$= - \int_h^{1-h} t^{-a} (1-t)^{a-1} dt + t^{-a} (1-t)^{a-1} dt = 0$$

$$g(a) = \text{constant} \quad \Rightarrow g(a) = \lambda$$

$$= - \int_h^{1-h} t^{-a} (1-t)^{a-1} dt + t^{-a} (1-t)^{a-1} dt = 0$$

$$= \int_h^{1-h} \frac{dt}{\sqrt{\left(t - \frac{1}{2}\right)^2 + \frac{1}{4}}} = \left(\sin^{-1} \frac{t - \frac{1}{2}}{\frac{1}{2}} \right) \Big|_h^{1-h} = \sin^{-1}(2t-1) \Big|_h^{1-h} = \sin^{-1}(1-2h) - \sin^{-1}(2h-1) = \pi$$

9. The value of $g'\left(\frac{1}{2}\right)$ is : [JEE(Adv.)2014]

$g'\left(\frac{1}{2}\right)$ का मान है :

[JEE(Adv.)2014]

- (A) $\pi/2$ (B) π (C) $-\pi/2$ (D) 0

Ans. D

Sol. $g(a) = \lim_{h \rightarrow 0^+} \int_h^{1-h} t^{-a} (1-t)^{a-1} dt$

$$g(1-a) = \lim_{h \rightarrow 0^+} \int_h^{1-h} t^{-(1-a)} (1-t)^{(1-a)-1} dt = \lim_{h \rightarrow 0^+} \int_h^{1-h} t^{a-1} (1-t)^{-a} dt$$

$$= \lim_{h \rightarrow 0^+} \int_h^{1-h} (1-t)^{a-1} (1-(1-t))^{-a} dt \left\{ \text{by } \int_a^b f(x) dx = \int_a^b f(a+b-x) dx \right\}$$

$$= \lim_{h \rightarrow 0^+} \int_h^{1-h} (1-t)^{a-1} t^{-a} dt$$

$$g(1-a) = g(a)$$

$$-g'(1-a) = g'(a)$$

$$\text{at } a = \frac{1}{2}$$

$$-g'\left(\frac{1}{2}\right) = g'\left(\frac{1}{2}\right) \Rightarrow g'\left(\frac{1}{2}\right) = 0$$

10. Match the following column :

[JEE(Adv.)2014]

Column-I

Column-II

P. The number of polynomials $f(x)$ with non-negative integer coefficients

1. 8

of degree ≤ 2 , satisfying $f(0) = 0$ and $\int_0^1 f(x) dx = 1$, is

Q. The number of points in the interval $[-\sqrt{13}, \sqrt{13}]$ at which

2. 2

$f(x) = \sin(x^2) + \cos(x^2)$ attains its maximum value, is

R. $\int_{-2}^2 \frac{3x^2}{(1+e^x)} dx$ equals

3. 4

S. $\frac{\left(\int_{-\frac{1}{2}}^{\frac{1}{2}} \cos 2x \log\left(\frac{1+x}{1-x}\right) dx\right)}{\left(\int_0^{\frac{1}{2}} \cos 2x \log\left(\frac{1+x}{1-x}\right) dx\right)}$ equals

4. 0

	P	Q	R	S
(A)	3	2	4	1
(C)	3	2	1	4

	P	Q	R	S
(B)	2	3	4	1
(D)	2	3	1	4

निम्न कॉलम को सुमेलित कीजिए :

[JEE(Adv.)2014]

Column-I

Column-II

P. अक्रेणात्मक पूर्णांक गुणांकों वाले बहुपदों $f(x)$, जिनकी घात

1. 8

≤ 2 है तथा जो $f(0) = 0$ एवं $\int_0^1 f(x) dx = 1$ को सन्तुष्ट करती है, की संख्या है

Q. अंतराल $[-\sqrt{13}, \sqrt{13}]$ में स्थित उन बिन्दुओं की संख्या जिन पर

2. 2

$f(x) = \sin(x^2) + \cos(x^2)$ का मान अधिकतम है,

R. $\int_{-2}^2 \frac{3x^2}{(1+e^x)} dx$ का मान है

3. 4

S. $\frac{\left(\int_{-\frac{1}{2}}^{\frac{1}{2}} \cos 2x \log \left(\frac{1+x}{1-x}\right) dx\right)}{\left(\int_0^{\frac{1}{2}} \cos 2x \log \left(\frac{1+x}{1-x}\right) dx\right)}$ का मान है

4. 0

	P	Q	R	S
(A)	3	2	4	1
(C)	3	2	1	4

	P	Q	R	S
(B)	2	3	4	1
(D)	2	3	1	4

Ans. D

Sol. (P) Let $f(x) = ax^2 + bx$, $a, b \in W$ (as $f(0) = 0$)

$$\int_1^0 (ax^2 + bx) dx = \frac{a}{3} + \frac{b}{2} = 1 \Rightarrow 2a + 3b = 6$$

$$(a, b) \equiv (3, 0), (0, 2)$$

$\Rightarrow$ Number of such polynomials = 2

$$(Q) f(x) = \sqrt{2} \sin \left(x^2 + \frac{\pi}{4} \right)$$

if $f(x)$ is maximum

$$x^2 + \frac{\pi}{4} = 2n\pi + \frac{\pi}{2}$$

$$x^2 = 2n\pi + \frac{\pi}{4}$$

$$\left\{ \because x \in [-\sqrt{13}, \sqrt{13}] \right\}$$

$$\Rightarrow n = 0 \text{ or } 1 \text{ or } -1 \text{ or } -2$$

$$\begin{aligned} \text{(R)} \int_{-2}^2 \frac{3x^2}{1+e^x} dx &= \int_0^2 3x^2 \left(\frac{1}{1+e^x} + \frac{1}{1+e^{-x}} \right) dx \left\{ \int_{-a}^a f(x) dx = \int_0^a (f(x) + f(-x)) dx \right\} \\ &= \int_0^2 3x^2 \left(\frac{1}{1+e^x} + \frac{e^x}{1+e^x} \right) dx \int_0^2 3x^2 dx = x^3 \Big|_0^2 = 8 \end{aligned}$$

$$\text{(S)} \int_{-1/2}^{1/2} \cos 2x \ln \left(\frac{1+x}{1-x} \right) dx = 0 \quad (\text{as it is an odd function})$$

Hence $P \rightarrow 2, Q \rightarrow 3, R \rightarrow 1, S \rightarrow 4$

11. Let $f: (0, \infty) \rightarrow \mathbb{R}$ be given by $f(x) = \int_{\frac{1}{x}}^x e^{-\left(t+\frac{1}{t}\right)} \frac{dt}{t}$. then : [JEE(Adv.)2014]

(A) $f(x)$ is monotonically increasing on $[1, \infty)$ (B) $f(x)$ is monotonically decreasing on $(0, 1)$

(C) $f(x) + f\left(\frac{1}{x}\right) = 0$, for all $x \in (0, \infty)$ (D) $f(2^x)$ is an odd function of x on $\mathbb{R}$

माना कि $f: (0, \infty) \rightarrow \mathbb{R}$ निम्न के द्वारा $f(x) = \int_{\frac{1}{x}}^x e^{-\left(t+\frac{1}{t}\right)} \frac{dt}{t}$ परिभाषित है। तब : [JEE(Adv.)2014]

(A) $[1, \infty)$ पर $f(x)$ एकदिष्ट वर्धमान है (B) $(0, 1)$ पर $f(x)$ एकदिष्ट ह्रसमान है

(C) सभी $x \in (0, \infty)$ के लिये, $f(x) + f\left(\frac{1}{x}\right) = 0$ (D) $\mathbb{R}$ पर $f(2^x)$, x का एक विषम फलन है

Ans. ACD

$$\text{Sol. } f(x) = \int_{\frac{1}{x}}^x e^{-\left(t+\frac{1}{t}\right)} \frac{dt}{t}$$

$$\Rightarrow f'(x) = \frac{e^{-\left(x+\frac{1}{x}\right)}}{x} + \frac{1}{x^2} e^{-\left(-x+\frac{1}{x}\right)} \cdot x$$

$$= e^{-\left(x+\frac{1}{x}\right)} \cdot \frac{2}{x}$$

$$\Rightarrow f'(x) > 0 \forall x \in (0, \infty)$$

$\Rightarrow f(x)$ is increasing in $(0, \infty)$

$\Rightarrow f(x)$ is increasing in $(0, \infty)$

$$\text{Now } f(x) + f\left(\frac{1}{x}\right) = \int_{\frac{1}{x}}^x \frac{e^{-\left(t+\frac{1}{t}\right)}}{t} dt + \int_x^{\frac{1}{x}} \frac{e^{-\left(t+\frac{1}{t}\right)}}{t} dt$$

$$= \int_{\frac{1}{x}}^x \frac{e^{-\left(t+\frac{1}{t}\right)}}{t} dt = 0$$

Similarly

$$f(2^x) + f\left(\frac{1}{2^x}\right) = 0$$

$$\Rightarrow f(2^{-x}) = -f(2^x)$$

12. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by $f(x) = \begin{cases} [x], & x \leq 2 \\ 0, & x > 2 \end{cases}$ where $[x]$ is the greatest integer less than or equal to

x. If $I = \int_{-1}^2 \frac{x f(x^2)}{2 + f(x+1)} dx$, then the value of $(4I - 1)$ is :

[JEE(Adv.)2015]

माना कि फलन $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \begin{cases} [x], & x \leq 2 \\ 0, & x > 2 \end{cases}$ से परिभाषित है, जहाँ $[x]$, x से कम या x के बराबर के महतम पूर्णांक को दर्शाता

है। यदि $I = \int_{-1}^2 \frac{x f(x^2)}{2 + f(x+1)} dx$, तब $(4I - 1)$ का मान है :

[JEE(Adv.)2015]

Ans. 0

Sol. $I = \int_{-1}^2 \frac{x [x^2]}{2 + [x+1]} dx = \int_{-1}^2 \frac{x [x^2]}{2 + [x+1]} dx = \int_{-1}^0 \frac{0}{2+0} dx + \int_0^1 \frac{0}{2+0} dx + \int_1^{\sqrt{2}} \frac{x \cdot 1}{2+2} dx$

$$= \frac{1}{4}$$

$$\therefore 4I - 1 = 0$$

13. If $\alpha = \int_0^1 \left(e^{9x+3 \tan^{-1} x} \right) \left(\frac{12+9x^2}{1+x^2} \right) dx$ where $\tan^{-1} x$ takes only principal values, then the value of $\left(\log_e |1+\alpha| - \frac{3\pi}{4} \right)$

is :

[JEE(Adv.)2015]

यदि $\alpha = \int_0^1 \left(e^{9x+3 \tan^{-1} x} \right) \left(\frac{12+9x^2}{1+x^2} \right) dx$ जहाँ $\tan^{-1} x$ केवल मुख्य मानों (principal values) को लेता है, तब

$\left(\log_e |1+\alpha| - \frac{3\pi}{4} \right)$ का मान है :

[JEE(Adv.)2015]

Ans. 9

Sol. $\alpha = \int_0^1 e^{9x+3 \tan^{-1} x} \cdot \left(\frac{12+9x^2}{1+x^2} \right) dx$

$$\Rightarrow \alpha = \left(e^{9x+3 \tan^{-1} x} \right)_0^1$$

$$\Rightarrow \alpha = e^{9+\frac{3\pi}{4}} - 1$$

$$\Rightarrow \ln(1+\alpha) = 9 + \frac{3\pi}{4}$$

Alter :

$$\alpha = \int_0^1 e^{(9x+3 \tan^{-1} x)} \left(\frac{12+9x^2}{1+x^2} \right) dx$$

Let $9x + 3 \tan^{-1} x = t$

$$\Rightarrow \left(9 + \frac{3}{1+x^2} \right) dx - dt \Rightarrow \left(\frac{12+9x^2}{1+x^2} \right) dx = dt$$

$$\Rightarrow \alpha = \int_0^{9+3\pi/4} e^t dt = \left(e^t \right)_0^{9+3\pi/4} = e^{9+3\pi/4} - 1$$

$$\text{Now } \log_e |1+\alpha| - 3\pi/4 = \log_e e^{(9+3\pi/4)} - 3\pi/4 = 9$$

14. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous odd function, which vanishes exactly at one point and $f(1) = \frac{1}{2}$. Suppose that

$$F(x) = \int_{-1}^x f(t) dt \text{ for all } x \in [-1, 2] \text{ and } G(x) = \int_{-1}^x t |f(f(t))| dt \text{ for all } x \in [-1, 2]. \text{ If } \lim_{x \rightarrow 1} \frac{F(x)}{G(x)} = \frac{1}{14}, \text{ then the}$$

value of $f\left(\frac{1}{2}\right)$ is :

[JEE(Adv.)2015]

माना कि $f: \mathbb{R} \rightarrow \mathbb{R}$ एक संतत विषम फलन है जिसका मान केवल एक बिन्दु पर ही शून्य होता है तथा $f(1) = \frac{1}{2}$ है। माना कि सभी x

$\in [-1, 2]$ के लिए $F(x) = \int_{-1}^x f(t) dt$ एवं सभी $x \in [-1, 2]$ के लिए $G(x) = \int_{-1}^x t |f(f(t))| dt$ हैं। यदि $\lim_{x \rightarrow 1} \frac{F(x)}{G(x)} = \frac{1}{14}$ है,

तब $f\left(\frac{1}{2}\right)$ का मान है :

[JEE(Adv.)2015]

Ans. 7

Sol. $F(x) = \int_{-1}^x f(t) dt = \int_1^x f(t) dt$

$$G(x) = \int_{-1}^x t |f(f(t))| dt = \int_{-1}^x t |f(f(t))| dt$$

$$\lim_{x \rightarrow 1} \frac{F(x)}{G(x)} = \frac{1}{14}$$

L'hospitals

$$\lim_{x \rightarrow 1} \frac{f(x)}{x |f(f(x))|} = \frac{1}{14}$$

$$\frac{\frac{1}{2}}{1 \left| f\left(\frac{1}{2}\right) \right|} = \frac{1}{14}$$

$$f\left(\frac{1}{2}\right) = 7$$

15. The option(s) with the values of a and L that satisfy the following equation is (are) $\frac{\int_0^{4\pi} e^t (\sin^6 at + \cos^4 at) dt}{\int_0^{\pi} e^t (\sin^6 at + \cos^4 at) dt} = L?$

[JEE(Adv.)2015]

निम्नलिखित में से a तथा L के कौन सा (से) मान समीकरण $\frac{\int_0^{4\pi} e^t (\sin^6 at + \cos^4 at) dt}{\int_0^{\pi} e^t (\sin^6 at + \cos^4 at) dt} = L?$ [JEE(Adv.)2015]

$$(A) a = 2, L = \frac{e^{4\pi} - 1}{e^\pi - 1} \quad (B) a = 2, L = \frac{e^{4\pi} + 1}{e^\pi + 1} \quad (C) a = 4, L = \frac{e^{4\pi} - 1}{e^\pi - 1} \quad (D) a = 4, L = \frac{e^{4\pi} + 1}{e^\pi + 1}$$

Ans. AC

$$\text{Sol. } I_1 = \int_0^\pi e^t (\sin^6 at + \cos^4 at) dt + \int_\pi^{2\pi} e^t (\sin^6 at + \cos^4 at) dt + \int_{2\pi}^{3\pi} e^t (\sin^6 at + \cos^4 at) dt$$

$$+ \int_{2\pi}^{4\pi} e^t (\sin^6 at + \cos^4 at) dt$$

$$= (1 + e^\pi + e^{2\pi} + e^{3\pi}) \int_\pi^{2\pi} e^t (\sin^6 at + \cos^4 at) dt$$

$$= (1 + e^\pi + e^{2\pi} + e^{3\pi}) \int_\pi^{2\pi} e^t (\sin^6 at + \cos^4 at) dt$$

$$\Rightarrow \frac{I_1}{I_2} = 1 + e^\pi + e^{2\pi} + e^{3\pi} = \frac{e^{4\pi} - 1}{e^\pi - 1}$$

16. Let $f(x) = 7 \tan^8 x + 7 \tan^6 x - 3 \tan^4 x - 3 \tan^2 x$ for all $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then the correct expression (s) is (are) :

[JEE(Adv.)2015]

माना कि सभी $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ के लिए, $f(x) = 7 \tan^8 x + 7 \tan^6 x - 3 \tan^4 x - 3 \tan^2 x$ है, तब सही कथन है (हैं) :

[JEE(Adv.)2015]

$$(A) \int_0^{\pi/4} x f(x) dx = \frac{1}{12} \quad (B) \int_0^{\pi/4} f(x) dx = 0 \quad (C) \int_0^{\pi/4} x f(x) dx = \frac{1}{6} \quad (D) \int_0^{\pi/4} f(x) dx = 1$$

Ans. AB

$$\text{Sol. } f(x) = (7 \tan^6 x - 3 \tan^2 x) \cdot \sec^2 x$$

$$\therefore \int_0^{\pi/4} f(x) dx = \int_0^1 (7t^6 - 3t^2) dt = (t^7 - t^3)_0^1 = 0$$

$$\text{Now } \int_0^{\pi/4} f(x) dx = \int_0^1 \underbrace{(7t^6 - 3t^2)}_I \underbrace{\tan^{-1} t}_I dt$$

$$= \left(\tan^{-1} t (t^7 - t^3) \right)_0^1 - \int_0^1 (t^7 - t^3) \cdot \frac{1}{1+t^2} dt$$

$$= \int_0^1 \frac{t^3(1-t^4)}{1+t^2} dt = \int_0^1 t^3(1-t^2) dt$$

$$= \frac{1}{4} - \frac{1}{6} = \frac{1}{12}$$

17. Let $f'(x) = \frac{192x^3}{2 + \sin^4 \pi x}$ for all $x \in \mathbb{R}$ with $f\left(\frac{1}{2}\right) = 0$. If $m \leq \int_{1/2}^1 f(x) dx \leq M$, then the possible values of m and M are : [JEE(Adv.)2015]

माना कि सभी $x \in \mathbb{R}$ के लिए, $f'(x) = \frac{192x^3}{2 + \sin^4 \pi x}$ एवं $f\left(\frac{1}{2}\right) = 0$ है। यदि $m \leq \int_{1/2}^1 f(x) dx \leq M$, तब m और M के सही संभव मान हैं (हैं) : [JEE(Adv.)2015]

(A) $m = 13, M = 24$ (B) $m = \frac{1}{4}, M = \frac{1}{2}$ (C) $m = -11, M = 0$ (D) $m = 1, M = 12$

Ans. D

Sol. $f'(x) = \frac{192x^3}{2 + \sin^4(\pi x)} \quad \forall x \in \mathbb{R}$

If $x \in \left[\frac{1}{2}, 1\right]$ then $f''(x) > 0$

$$\Rightarrow f'\left(\frac{1}{2}\right) \leq f'(x) \leq f'(1)$$

$$\Rightarrow \frac{24}{3} \leq f'(x) \leq \frac{192}{2}$$

$$8 \leq f'(x) \leq 96$$

$$\int_{1/2}^x 8 dx \leq \int_{1/2}^x f'(x) dx \leq \int_{1/2}^x 96$$

$$8\left(x - \frac{1}{2}\right) \leq f(x) - f\left(\frac{1}{2}\right) \leq 96\left(x - \frac{1}{2}\right)$$

$$8x - 4 \leq f(x) \leq 96x - 48$$

$$\int_{1/2}^1 (8x - 4) dx \leq \int_{1/2}^1 f(x) dx \leq \int_{1/2}^1 (96x - 48) dx$$

$$\left[4x^2 - 4x\right]_{1/2}^1 \leq \int_{1/2}^1 f(x) dx \leq \left[48x^2 - 48x\right]_{1/2}^1$$

$$1 \leq \int_{1/2}^1 f(x) dx \leq 12$$

$$\Rightarrow m = 1 \text{ \& } M = 12$$

Paragraph for Question Nos. 18 to 19

Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be a thrice differentiable function. Suppose that $F(1) = 0$, $F(3) = -4$ and $F'(x) < 0$ for all $x \in (1/2, 3)$. Let $f(x) = xF(x)$ for all $x \in \mathbb{R}$.

माना कि $F : \mathbb{R} \rightarrow \mathbb{R}$ एक फलन है जो तीन बार अवकलनीय (thrice differentiable) है। माना कि $F(1) = 0$, $F(3) = -4$ और सभी $x \in (1/2, 3)$ के लिए, $F'(x) < 0$ है। माना कि सभी $x \in \mathbb{R}$ के लिए, $f(x) = xF(x)$ है।

18. The correct statement (s) is (are) :

[JEE(Adv.)2015]

(A) $f'(1) < 0$

(B) $f(2) < 0$

(C) $f'(x) \neq 0$ for any $x \in (1, 3)$

(D) $f'(x) = 0$ for some $x \in (1, 3)$

निम्नलिखित में से सही कथन है (हैं) :

[JEE(Adv.)2015]

(A) $f'(1) < 0$

(B) $f(2) < 0$

(C) किसी भी $x \in (1, 3)$ के लिए $f'(x) \neq 0$

(D) कुछ $x \in (1, 3)$ के लिए $f'(x) = 0$

Ans. ABC

Sol. $f'(x) = x F'(x) + F(x)$

$\Rightarrow f'(1) = F'(1) + F(1) = F'(1) < 0 \Rightarrow (A)$

$f(2) = 2F(2) < 0 \Rightarrow (B)$

for $x \in (1, 3)$ $f'(x) = xF'(x) + F(x) < 0 \Rightarrow (C)$

19. If $\int_1^3 x^2 F'(x) dx = -12$ and $\int_1^3 x^3 F''(x) dx = 40$, then the correct expression (s) is (are) :

[JEE(Adv.)2015]

यदि $\int_1^3 x^2 F'(x) dx = -12$ और $\int_1^3 x^3 F''(x) dx = 40$ है, तब सही कथन है (हैं) :

[JEE(Adv.)2015]

(A) $9f'(3) + f'(1) - 32 = 0$

(B) $\int_1^3 f(x) dx = 12$

(C) $9f'(3) - f'(1) + 32 = 0$

(D) $\int_1^3 f(x) dx = -12$

Ans. CD

Sol. $\int_1^3 x^2 F'(x) dx = -12$

$\Rightarrow \left[x^2 F(x) \right]_1^3 - 2 \int_1^3 x F(x) dx = -12$

$$\Rightarrow [x f(x)]_1^3 - 2 \int_1^3 f(x) dx = -12$$

$$\Rightarrow \int_1^3 f(x) dx = -12$$

$$\text{Now } \int_1^3 x^3 F''(x) dx = [x^3 F'(x)]_1^3 - 3 \int_1^3 x^2 F'(x) dx = 40$$

$$\Rightarrow 27F'(3) - F'(1) + 36 = 40$$

$$9f'(3) - f'(1) + 32 = 0$$

20. The total number of distinct $x \in [0, 1]$ for which $\int_0^x \frac{t^2}{1+t^4} dt = 2x - 1$ is.

[JEE(Adv.)2016]

ऐसे सभी भिन्न $x \in [0, 1]$, जिनके लिए $\int_0^x \frac{t^2}{1+t^4} dt = 2x - 1$ है, की कुल संख्या है।

[JEE(Adv.)2016]

Ans. 1

Sol. Let $f(x) = \int_0^x \frac{t^2}{1+t^4} dt$

Differentiating both sides, we get

$$f'(x) = \frac{x^2}{1+x^4}$$

Observing $f'(x)$, we can say that, $f'(x) > 0$

$\Rightarrow f(x)$ is an increasing function

$$\text{Now at } x = 0, f(0) = 0 \text{ and at } x = 1, f(1) = \int_0^1 \frac{t^2}{1+t^4} dt$$

$$\text{Because, } 0 < \frac{x^2}{1+x^4} < 1$$

$$\Rightarrow 0 < \int_0^1 \frac{t^2}{1+t^4} dt < \int_0^1 1 dt$$

$$\Rightarrow 0 < \int_0^1 \frac{t^2}{1+t^2} dt < 1$$

Plotting graph of $y = f(x)$ and $y = 2x - 1$, we find that they intersect at only one point

21. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{x^2 \cos x}{1+e^x} dx$ is equal to : [JEE(Adv.)2016]

$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{x^2 \cos x}{1+e^x} dx$ का मान है : [JEE(Adv.)2016]

- (A) $\frac{\pi^2}{4} - 2$ (B) $\frac{\pi^2}{4} + 2$ (C) $\pi^2 - e^{\frac{\pi}{2}}$ (D) $\pi^2 + e^{\frac{\pi}{2}}$

Ans. A

Sol. Apply $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx$

$$\text{Let } I = \int_{-\pi/2}^{\pi/2} \frac{x^2 \cos x}{1+e^x} dx \Rightarrow 2I = \int_{-\pi/2}^{\pi/2} \frac{x^2 \cos x}{1+e^x} dx + \int_{-\pi/2}^{\pi/2} \frac{x^2 \cos x}{1+e^{-x}} dx$$

$$I = \int_0^{\pi/2} x^2 \cos x dx = [x^2 \sin x - 2x(-\cos x) + 2(-\sin x)]_0^{\pi/2} = \frac{\pi^2}{4} - 2$$

22. Let $f(x) = \lim_{n \rightarrow \infty} \left(\frac{n^n (x+n)(x+\frac{n}{2}) \dots (x+\frac{n}{n})}{n!(x^2+n^2)(x^2+\frac{n^2}{4}) \dots (x^2+\frac{n^2}{n^2})} \right)^{\frac{x}{n}}$, for all $x > 0$. Then : [JEE(Adv.)2016]

माना कि सभी $x > 0$ के लिए $f(x) = \lim_{n \rightarrow \infty} \left(\frac{n^n (x+n)(x+\frac{n}{2}) \dots (x+\frac{n}{n})}{n!(x^2+n^2)(x^2+\frac{n^2}{4}) \dots (x^2+\frac{n^2}{n^2})} \right)^{\frac{x}{n}}$ है। तब : [JEE(Adv.)2016]

- (A) $f\left(\frac{1}{2}\right) \geq f(1)$ (B) $f\left(\frac{1}{3}\right) \leq f\left(\frac{2}{3}\right)$ (C) $f'(2) \leq 0$ (D) $\frac{f'(3)}{f(3)} \geq \frac{f'(2)}{f(2)}$

Ans. BC

Sol.
$$f(x) = \lim_{n \rightarrow \infty} \left(\frac{\prod_{r=1}^n \left(x + \frac{n}{r} \right)}{\prod_{r=1}^n \left(\frac{r}{n} \right) \prod_{r=1}^n \left(x^2 + \frac{n^2}{r^2} \right)} \right)^{\frac{x}{n}}$$

$$\Rightarrow \ln f(x) = x \lim_{n \rightarrow \infty} \left[\left\{ \sum_{r=1}^n \log \left(x + \frac{1}{r/n} \right) \cdot \frac{1}{n} \right\} - \left\{ \sum_{r=1}^n \log \left(\frac{r}{n} \right) \cdot \frac{1}{n} \right\} - \left\{ \sum_{r=1}^n \log \left(x^2 + \frac{1}{(r/n)^2} \right) \cdot \frac{1}{n} \right\} \right]$$

$$\Rightarrow \ln f(x) = x \int_0^1 \log \left(x + \frac{1}{t} \right) dt - \int_0^1 \log(t) dt - \int_0^1 \log \left(x^2 + \frac{1}{t^2} \right) dt$$

$$\Rightarrow \ln f(x) = x \int_0^1 \log \left(\frac{tx+1}{t^2x^2+1} \right) dt$$

put $tx = u$

$$\Rightarrow \ln f(x) = \int_0^x \log \left(\frac{u+1}{u^2+1} \right) du$$

$$\Rightarrow f(x) = \int_0^x \log \left(\frac{u+1}{u^2+1} \right) du$$

23. If $I = \sum_{k=1}^{98} \int_k^{k+1} \frac{k+1}{x(x+1)} dx$, then :

[JEE(Adv.)2017]

यदि $I = \sum_{k=1}^{98} \int_k^{k+1} \frac{k+1}{x(x+1)} dx$, तब :

[JEE(Adv.)2017]

(A) $I > \frac{49}{50}$ (B) $I < \frac{49}{50}$ (C) $I < \log_e 99$ (D) $I > \log_e 99$

Ans. AC

Sol.
$$I = \sum_{k=1}^{98} \left(\int_k^{k+1} \frac{k+1}{x(x+1)} dx \right)$$

For $\int_k^{k+1} \frac{k+1}{x(x+1)} dx$

$$\frac{k+1}{x(x+1)} < \frac{k+1}{x(k+1)} \quad (\because \text{Least value of } x+1 \text{ is } k+1)$$

$$\Rightarrow \frac{k+1}{x(x+1)} < \frac{1}{x}$$

$$x < k+1$$

$$\frac{k+1}{x} > 1$$

$$\Rightarrow \frac{k+1}{x(x+1)} > \frac{1}{x+1}$$

$$\Rightarrow \frac{1}{x+1} < \frac{k+1}{x(x+1)} < \frac{1}{x}$$

$$\Rightarrow \int_k^{k+1} \frac{1}{x+1} dx < \int_k^{k+1} \frac{k+1}{x(x+1)} dx < \int_k^{k+1} \frac{1}{x} dx$$

$$\Rightarrow [\ln(x+1)]_k^{k+1} < \int_k^{k+1} \frac{k+1}{x(x+1)} dx < (\ln x)_k^{k+1}$$

$$\sum_{k=1}^{98} \ln\left(\frac{k+2}{k+1}\right) < I < \sum_{k=1}^{98} \ln\left(\frac{k+1}{k}\right)$$

$$\sum_{k=1}^{98} \ln\left(\frac{k+2}{k+1}\right) = (\ln 3 - \ln 2 + \ln 4 - \ln 3 + \dots + \ln 100 - \ln 99)$$

$$= \ln\left(\frac{100}{2}\right) = \ln 50$$

$$\sum_{k=1}^{98} \ln\left(\frac{k+1}{k}\right) = \sum_{k=1}^{98} (\ln(k+1) - \ln k)$$

$$= \ln 2 - \ln 1 + \ln 3 - \ln 2 + \dots + \ln 99 - \ln 98$$

$$= \ln(99)$$

$$\Rightarrow \ln(50) < I < \ln(99)$$

$$\text{as } \ln(50) > \frac{49}{50}$$

$$\Rightarrow I > \frac{49}{50} \text{ \& } I < \ln(99)$$

24. If $g(x) = \int_{\sin x}^{\sin(2x)} \sin^{-1}(t) dt$, then :

[JEE(Adv.)2017]

यदि $g(x) = \int_{\sin x}^{\sin(2x)} \sin^{-1}(t) dt$ है, तब :

[JEE(Adv.)2017]

$$(A) g'\left(-\frac{\pi}{2}\right) = -2\pi \quad (B) g'\left(-\frac{\pi}{2}\right) = 2\pi \quad (C) g'\left(\frac{\pi}{2}\right) = 2\pi \quad (D) g'\left(\frac{\pi}{2}\right) = -2\pi$$

Ans. BONUS

Sol. $g(x) = \int_{\sin x}^{\sin 2x} \sin^{-1}(t) dt$

$$g'(x) = \sin^{-1}(\sin 2x) \cdot \cos 2x \cdot 2 - \sin^{-1}(\sin x) \cdot \cos x$$

$$= 2\cos 2x \cdot \sin^{-1}(\sin 2x) - \cos x \cdot \sin^{-1}(\sin x)$$

$$g'\left(-\frac{\pi}{2}\right) = \cos(-\pi) \sin^{-1}(\sin(-\pi)) - \cos\left(-\frac{\pi}{2}\right) \cdot \sin^{-1}\left(\sin\left(-\frac{\pi}{2}\right)\right) = 0$$

$$g'\left(-\frac{\pi}{2}\right) = 2\cos(\pi) \sin^{-1}(\sin(\pi)) - \cos\left(\frac{\pi}{2}\right) \cdot \sin^{-1}\left(\sin\left(\frac{\pi}{2}\right)\right) = 0$$

25. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that $f(0) = 0$, $f\left(\frac{\pi}{2}\right) = 3$ and $f'(0) = 1$.

If $g(x) = \int_x^{\frac{\pi}{2}} [f'(t) \operatorname{cosec} t - \cot t \operatorname{cosec} t f(t)] dt$ for $x \in (0, \frac{\pi}{2}]$, then $\lim_{x \rightarrow 0} g(x) =$ **[JEE(Adv.)2017]**

माना $f: \mathbb{R} \rightarrow \mathbb{R}$ इस प्रकार का अवकलनीय फलन (differentiable function) है, कि $f(0) = 0$, $f\left(\frac{\pi}{2}\right) = 3$ तथा $f'(0) = 1$ है।

यदि $x \in (0, \frac{\pi}{2}]$ के लिये $g(x) = \int_x^{\frac{\pi}{2}} [f'(t) \operatorname{cosec} t - \cot t \operatorname{cosec} t f(t)] dt$ है, तब $\lim_{x \rightarrow 0} g(x) =$ **[JEE(Adv.)2017]**

Ans. 2

Sol. $g(x) = [f(t) \operatorname{cosec} t]_0^{\pi/2}$

$$= f(\pi/2) - f(x) \operatorname{cosec} x$$

$$g(x) = 3 - \frac{f(x)}{\sin x}$$

$$\lim_{x \rightarrow 0} g(x) = 3 - \lim_{x \rightarrow 0} \frac{f(x)}{\sin x}$$

$$= 3 - \lim_{x \rightarrow 0} \frac{f'(x)}{\cos x} \quad (\text{Use L.H. rule})$$

$$= 3 - 1 = 2$$

26. Let $f: \mathbb{R} \rightarrow (0, 1)$ be a continuous function. Then, which of the following function(s) has(have) the value zero at some point in the interval $(0, 1)$? **[JEE(Adv.)2017]**

माना कि $f: \mathbb{R} \rightarrow (0, 1)$ एक सतत फलन है। तब निम्न फलनों में से कौनसे फलन/फलनों का/के मान अन्तराल $(0, 1)$ के किसी बिन्दु पर शून्य होगा? :

[JEE(Adv.)2017]

(A) $x^9 - f(x)$ (B) $f(x) + \int_0^{\frac{\pi}{2}} f(t) \sin t \, dt$ (C) $x - \int_0^{\frac{\pi}{2}-x} f(t) \cos t \, dt$ (D) $e^x - \int_0^x f(t) \sin t \, dt$

Ans. AC

Sol. Let $g(x) = x^9 - f(x)$

$$\Rightarrow g(0) = -f(0) < 0 \text{ and } g(1) = 1 - f(1) > 0$$

$$\Rightarrow g(x) = 0 \text{ for some } x \in (0, 1)$$

$$\text{Let } h(x) = x - \int_0^{\frac{\pi}{2}-x} f(t) \cos t \, dt$$

$$h(0) = 0 - \int_0^{\frac{\pi}{2}-0} f(t) \cos t \, dt < 0$$

$$h(1) = 1 - \int_0^{\frac{\pi}{2}-1} f(t) \cos t \, dt > 0$$

$$\Rightarrow h(x) = 0 \text{ for some } x \in (0, 1)$$

$$e^x - \int_0^x f(t) \sin t \, dt$$

$$\because x \in (0, 1) \Rightarrow e^x \in (1, e)$$

$$\text{and } 0 < f(t) < 1 \text{ and } 0 < \sin t < 1$$

$$e^x - \int_0^x f(t) \sin t \, dt \neq 0$$

$$f(x) + \int_0^{\frac{\pi}{2}} f(t) \sin t \, dt > 0$$

27. The value of the integral $\int_0^{\frac{1}{2}} \frac{1 + \sqrt{3}}{\left((x+1)^2 (1-x)^6\right)^{1/4}} dx$ is :

[JEE(Adv.)2018]

समाकल $\int_0^{\frac{1}{2}} \frac{1 + \sqrt{3}}{\left((x+1)^2 (1-x)^6\right)^{1/4}} dx$ का मान है :

[JEE(Adv.)2018]

Ans. 2

Sol. $\frac{1 + \sqrt{3}}{\left((x+1)^2 (1-x)^6\right)^{1/4}}$

$$\int_0^{1/2} \frac{(1+\sqrt{3})dx}{\left[(1+x)^2(1-x)^6\right]^{1/4}}$$

$$\text{Put } \frac{1-x}{1+x} = t \Rightarrow \frac{-2dx}{(1+x)^2} = dt$$

$$I = \int_0^{1/3} \frac{(1+\sqrt{3})dt}{-2t^{6/4}} = \frac{-(1+\sqrt{3})}{2} \times \left| \frac{-2}{\sqrt{t}} \right|_1^{1/3} = (1+\sqrt{3})(\sqrt{3}-1) = 2$$

28. For each positive integer n , let $y_n = \frac{1}{n}((n+1)(n+2)\dots(n+n))^{\frac{1}{n}}$. For $x \in \mathbb{R}$, let $[x]$ be the greatest integer less than or equal to x . If $\lim_{n \rightarrow \infty} y_n = L$, then the value of $[L]$ is _____.

[JEE(Adv.)2018]

प्रत्येक धनात्मक पूर्णांक n के लिये, माना कि $y_n = \frac{1}{n}((n+1)(n+2)\dots(n+n))^{\frac{1}{n}}$.

$x \in \mathbb{R}$ के लिये माना कि $[x]$, x से छोटा या x के बराबर महत्तम पूर्णांक है। यदि $\lim_{n \rightarrow \infty} y_n = L$, तब $[L]$ का मान है _____

[JEE(Adv.)2018]

Ans. 1

$$\text{Sol. } y_n = \left(\frac{n+1}{n} \frac{n+2}{n} \dots \frac{n+n}{n} \right)^{1/n}$$

$$\log L = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \log \left(1 + \frac{r}{n} \right)$$

$$= \int_0^1 \log(1+x) dx = \int_1^2 \log x dx = [x \log x - x]_1^2 = 2 \log 2 = \log \frac{4}{3}$$

$$\Rightarrow L = \frac{4}{e} \Rightarrow [L] = 1$$

29. For $a \in \mathbb{R}$, $|a| > 1$, Let $\lim_{n \rightarrow \infty} \left(\frac{1 + \sqrt[3]{2} + \dots + \sqrt[3]{n}}{n^{7/3} \left(\frac{1}{(an+1)^2} + \frac{1}{(an+2)^2} + \dots + \frac{1}{(an+n)^2} \right)} \right) = 54$. Then the possible value (s) of

a is/are :

[JEE(Adv.)2018]

माना कि $a \in \mathbb{R}$, $|a| > 1$ के लिए $\lim_{n \rightarrow \infty} \left(\frac{1 + \sqrt[3]{2} + \dots + \sqrt[3]{n}}{n^{7/3} \left(\frac{1}{(an+1)^2} + \frac{1}{(an+2)^2} + \dots + \frac{1}{(an+n)^2} \right)} \right) = 54$ तब a का (के) सम्भावित मान है (हैं)

(A) -6 (B) 8 (C) 7 (D) -9

[JEE(Adv.)2018]

Ans. BD

Sol. $\lim_{x \rightarrow \infty} \left(\frac{1 + (2)^{1/3} + (3)^{1/3} \dots (n)^{1/3}}{n^{1/3} \cdot \frac{n^2}{n^2} \left(\frac{1}{\left(a + \frac{1}{n}\right)^2} + \frac{1}{\left(a + \frac{2}{n}\right)^2} \dots \frac{1}{\left(a + \frac{n}{n}\right)^2} \right)} \right)$

$$\lim_{x \rightarrow \infty} \left(\frac{\frac{1}{n} \sum_{r=1}^n \left(\frac{r}{n}\right)^{1/3}}{\frac{1}{n} \sum_{r=1}^n \frac{1}{\left(a + \frac{r}{n}\right)^2}} \right)$$

$$\frac{r}{n} = x, \frac{1}{n} = dx$$

$$\frac{\int_0^1 (x)^{1/3} dx}{\int_0^1 \frac{1}{(a+x)^2} dx} = 54$$

$$\frac{\left(\frac{x^{4/3}}{4/3}\right)_0^1}{-\left(\frac{1}{a+x}\right)_0^1} = 54$$

$$\frac{\frac{3}{4}}{-\left[\frac{1}{a+1} - \frac{1}{a}\right]} = 54$$

$$\frac{a(a+1)}{-(a-a-1)} = 72$$

$$a^2 + a - 72 = 0$$

$$a^2 + 9a - 8a - 72 = 0$$

$$a = -9, a = 8$$

30. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = (x-1)(x-2)(x-5)$ define $F(x) = \int_0^x f(t) dt, x > 0$. Then which of the following

options is/are correct :

[JEE(Adv.)2019]

(A) $F(x) \neq 0$ for all $x \in (0,5)$

(B) F has a local maximum at $x = 2$

(C) F has a local minimum at $x = 1$

(D) F has two local maxima and one local minimum in $(0, \infty)$

माना कि $f: \mathbb{R} \rightarrow \mathbb{R}, f(x) = (x-1)(x-2)(x-5)$ द्वारा दिया गया है। परिभाषित कीजिए। $F(x) = \int_0^x f(t) dt, x > 0$ तब निम्न में

से कौन सा (से) विकल्प सही है (हैं)?

[JEE(Adv.)2019]

(A) सभी $x \in (0,5)$ के लिए $F(x) \neq 0$

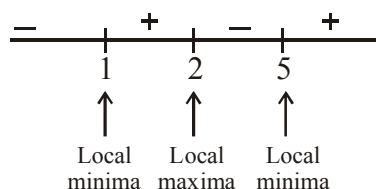
(B) F का एक स्थानीय उच्चतम (local maximum) $x = 2$ है

(C) F का एक स्थानीय निम्नतम (local minimum) $x = 1$ पर है

(D) F के दो स्थानीय उच्चतम और एक स्थानीय निम्नतम $(0, \infty)$ में हैं

Ans. ABC

Sol. $F'(x) = f(x) = (x-1)(x-2)(x-5)$



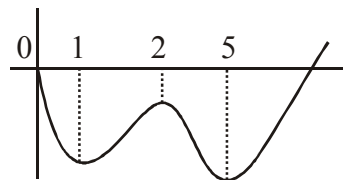
$$F(x) = \int_0^x (x^3 - 8x^2 + 17x - 10) dx$$

$$F(x) = \frac{x^4}{4} - \frac{8x^3}{3} + \frac{17}{2}x^2 - 10x$$

$$F(1) = \frac{1}{4} - \frac{8}{3} + \frac{17}{2} - 10 < 0$$

$$F(2) = \frac{16}{4} - \frac{64}{3} + \frac{68}{2} - 20 < 0$$

$$F(5) = \frac{625}{4} - \frac{1000}{3} + \frac{17 \times 25}{2} - 50 < 0$$



31. If $I = \frac{2}{\pi} \int_{-\pi/4}^{\pi/4} \frac{dx}{(1 + e^{\sin x})(2 - \cos 2x)}$ then $27I^2$ equals.

[JEE(Adv.)2019]

यदि $I = \frac{2}{\pi} \int_{-\pi/4}^{\pi/4} \frac{dx}{(1 + e^{\sin x})(2 - \cos 2x)}$ तब $27I^2$ बराबर ।

[JEE(Adv.)2019]

Ans. 4

Sol. Apply king property and add

$$2I = \frac{2}{\pi} \int_{-\pi/4}^{\pi/4} \frac{dx}{(2 - \cos 2x)}$$

$$\Rightarrow I = \frac{2}{\pi} \int_0^{\pi/4} \frac{dx}{2 - \left(\frac{1 - \tan^2 x}{1 + \tan^2 x} \right)}$$

let $\tan x = t$

$$\sec^2 x \, dx = dt$$

$$I = \frac{2}{\pi} \int_0^1 \frac{dt}{2 + 2t^2 - 1 + t^2}$$

$$I = \frac{2}{\pi} \int_0^1 \frac{dt}{1 + 3t^2} = \frac{2}{3\pi} \int_0^1 \frac{dt}{t^2 + \left(\frac{1}{\sqrt{3}} \right)^2}$$

$$= \frac{2}{3\pi} \cdot \frac{1}{\frac{1}{\sqrt{3}}} \left(\tan^{-1}(\sqrt{3}t) \right)_0^1$$

$$I = \frac{2}{\pi\sqrt{3}} \cdot \frac{\pi}{3} = \frac{2}{3\sqrt{3}}$$

$$I^2 = \frac{4}{27}$$

32. The value of the integral $\int_0^{\pi/2} \frac{3\sqrt{\cos \theta}}{(\sqrt{\cos \theta} + \sqrt{\sin \theta})^5} d\theta$ equals _____. [JEE(Adv.)2019]

समाकल (integral) $\int_0^{\pi/2} \frac{3\sqrt{\cos \theta}}{(\sqrt{\cos \theta} + \sqrt{\sin \theta})^5} d\theta$ का मान बराबर _____ [JEE(Adv.)2019]

Ans. 0.5

Sol. $\int_0^{\pi/2} \frac{3\sqrt{\cos \theta}}{(\sqrt{\cos \theta} + \sqrt{\sin \theta})^5} d\theta$

$$\int_0^{\pi/2} \frac{3 \sec^2 \theta}{(1 + \sqrt{\tan \theta})^5} d\theta$$

Let $\tan \theta = t^2$

$$\int_0^{\infty} \frac{6t dt}{(1+t)^5}$$

$\sec^2 \theta d\theta = 2t dt$

$$= 6 \left[\int_0^{\infty} \left(\frac{1}{(1+t)^4} - \frac{1}{(1+t)^5} \right) dt \right]$$

$$= 6 \left(-\frac{1}{3} \frac{1}{(1+t)^3} + \frac{1}{4(1+t)^4} \right)_0^{\infty}$$

$$= 6 \left(\frac{1}{3} - \frac{1}{4} \right)$$

$$= \frac{6}{12} = 0.5$$

33. Which of the following inequalities is/are TRUE : [JEE(Adv.)2020]

निम्न में से कौनसी असमिका(यें) सत्य है/हैं :

[JEE(Adv.)2020]

(A) $\int_0^1 x \cos x \, dx \geq \frac{3}{8}$ (B) $\int_0^1 x \sin x \, dx \geq \frac{3}{10}$ (C) $\int_0^1 x^2 \cos x \, dx \geq \frac{1}{2}$ (D) $\int_0^1 x^2 \sin x \, dx \geq \frac{2}{9}$

Ans. ABD

Sol. $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$

$$\Rightarrow \cos x \geq 1 - \frac{x^2}{2}$$

$$\begin{aligned} \text{(A)} \quad \int_0^1 x \cos x \, dx &\geq \int_0^1 x \left(1 - \frac{x^2}{2}\right) dx \\ &\geq \left(\frac{x^2}{2} - \frac{x^4}{8}\right)_0^1 \\ &\geq \frac{3}{8} \end{aligned}$$

$$\begin{aligned} \text{(B)} \quad \sin x &\geq x - \frac{x^3}{3!} \\ \int_0^1 x \sin x \, dx &\geq \int_0^1 \left(x^2 - \frac{x^4}{6}\right) dx \\ &\geq \left(\frac{x^3}{3} - \frac{x^5}{30}\right)_0^1 \\ &\geq \frac{3}{10} \end{aligned}$$

(C) $\cos x \leq 1$

$$\begin{aligned} \int_0^1 x^2 \cos x \, dx &\leq \int_0^1 x^2 \, dx \\ &\leq \left(\frac{x^3}{3}\right)_0^1 \\ &\leq \frac{1}{3} \end{aligned}$$

(D) $\sin x \geq x - \frac{x^3}{3!}$

$$\int_0^1 x^2 \sin x \, dx \geq \int_0^1 \left(x^3 - \frac{x^5}{6}\right) dx$$

$$\geq \left(\frac{x^4}{4} - \frac{x^6}{36} \right)_0^1$$

$$\geq \frac{2}{9}$$

34. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that its derivative f' is continuous and $f(\pi) = -6$.

If $F : [0, \pi] \rightarrow \mathbb{R}$ is defined by $F(x) = \int_0^x f(t) dt$, and if $\int_0^\pi (f'(x) + F(x)) \cos x dx = 2$, then the value of $f(0)$

is _____

[JEE(Adv.)2020]

माना कि $f : \mathbb{R} \rightarrow \mathbb{R}$ एक ऐसा अवकलनीय फलन है जिसका अवकलज f' संतत है, एवं $f(\pi) = -6$ है। यदि $F : [0, \pi] \rightarrow \mathbb{R}$,

$F(x) = \int_0^x f(t) dt$ द्वारा परभाषित है, तथा यदि $\int_0^\pi (f'(x) + F(x)) \cos x dx = 2$, तब $f(0)$ का मान है _____

[JEE(Adv.)2020]

Ans. 4

Sol. $F(x) = \int_0^x f(t) dt$

differentiating wrt x

$$F'(x) = f(x) \dots (1)$$

$$\int_0^\pi (f'(x) \cos x + F(x) \cos x) dx = 2$$

$$\Rightarrow \int_0^\pi f'(x) \cos x dx + \int_0^\pi F(x) \cos x dx = 2$$

Using integration by parts

$$[f(x) \cos x]_0^\pi + \int_0^\pi \sin x \cdot f(x) dx + [F(x) \cdot \sin x]_0^\pi - \int_0^\pi \sin x F'(x) dx = 2$$

Using (1)

$$[f(x) \cos x]_0^\pi + \int_0^\pi \sin x \cdot f(x) dx + [F(x) \cdot \sin x]_0^\pi - \int_0^\pi \sin x f(x) dx = 2$$

$$\Rightarrow -f(\pi) - f(0) + F(\pi) \cdot 0 - F(0) \cdot 0 = 2$$

$$\Rightarrow -f(\pi) - f(0) = 2$$

$$\Rightarrow 6 - f(0) = 2$$

$$\Rightarrow f(0) = 4 \text{ Ans.}$$

35. Let $f : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow \mathbb{R}$ be a continuous function such that $f(0) = 1$ and $\int_0^{\frac{\pi}{3}} f(t) dt = 0$. The which of the following statements is (are) **TRUE**? **[JEE(Adv.)2021]**

(A) The equation $f(x) - 3 \cos 3x = 0$ has at least one solution in $\left(0, \frac{\pi}{3}\right)$

(B) The equation $f(x) - 3 \sin 3x = -\frac{6}{\pi}$ has at least one solution in $\left(0, \frac{\pi}{3}\right)$

(C) $\lim_{x \rightarrow 0} \frac{x \int_0^x f(t) dt}{1 - e^{x^2}} = 1$

(D) $\lim_{x \rightarrow 0} \frac{\sin x \int_0^x f(t) dt}{x^2} = -1$

माना $f : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow \mathbb{R}$ एक सतत् फलन है जबकि $f(0) = 1$ तथा $\int_0^{\frac{\pi}{3}} f(t) dt = 0$ तब निम्न में से कौनसा/कौनसे कथन सही है/हैं?

[JEE(Adv.)2021]

(A) $\left(0, \frac{\pi}{3}\right)$ में समीकरण $f(x) - 3 \cos 3x = 0$ का कम से कम एक हल है

(B) $\left(0, \frac{\pi}{3}\right)$ में समीकरण $f(x) - 3 \sin 3x = -\frac{6}{\pi}$ का कम से कम एक हल है

(C) $\lim_{x \rightarrow 0} \frac{x \int_0^x f(t) dt}{1 - e^{x^2}} = -1$

(D) $\lim_{x \rightarrow 0} \frac{\sin x \int_0^x f(t) dt}{x^2} = -1$

Ans. ABC

Sol. (A) $\int_0^{\frac{\pi}{3}} (f(x) - 3 \cos 3x) dx = \int_0^{\frac{\pi}{3}} f(x) dx - \int_0^{\frac{\pi}{3}} 3 \cos 3x dx = 0 - \sin 3x \Big|_0^{\pi/3} = 0$

$$\Rightarrow f(x) - 3 \cos 3x = 0 \text{ has atleast one solution in } \left(0, \frac{\pi}{3}\right)$$

$$(B) \int_0^{\frac{\pi}{3}} \left(f(x) - 3 \sin 3x + \frac{6}{\pi} \right) dx = \int_0^{\frac{\pi}{3}} f(x) dx - 3 \int_0^{\frac{\pi}{3}} \sin 3x dx + \frac{6}{\pi} \int_0^{\frac{\pi}{3}} dx = 2 - 2 = 0$$

$\Rightarrow f(x) - 3 \sin 3x + \frac{6}{\pi} = 0$ has atleast one solution in $\left(0, \frac{\pi}{3}\right)$.

$$(C) \lim_{x \rightarrow \infty} \frac{x \int_0^x f(t) dt}{1 - e^{x^2}} \left(\frac{0}{0} \right)$$

Use L'hospital Rule

$$\lim_{x \rightarrow 0} \frac{x \cdot f(x) + \int_0^x f(t) dt}{-2xe^{x^2}} \left(\frac{0}{0} \right)$$

Use L'Hospital Rule again

$$\lim_{x \rightarrow 0} \frac{x \cdot f'(x) + 2f(x)}{-2e^{x^2} - 4x^2 \cdot e^{x^2}} = \frac{2 \cdot f(0)}{-2} = -1$$

$$(D) \lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right) \cdot \frac{\int_0^x f(t) dt}{x}$$

$$\lim_{x \rightarrow 0} \frac{\int_0^x f(t) dt}{x} \left(\frac{0}{0} \right)$$

Use L'Hospital Rule

$$\lim_{x \rightarrow 0} \frac{f(x)}{1} = f(0) = 1$$

As ABC

Question Stem for Question Nos. 36 and 37

Let $g_i : \left[\frac{\pi}{8}, \frac{3\pi}{8} \right] \rightarrow R, i = 1, 2$, and $f : \left[\frac{\pi}{8}, \frac{3\pi}{8} \right] \rightarrow R$ be functions such that

$$g_1(x) = 1, g_2(x) = |4x - \pi| \text{ and } f(x) = \sin^2 x, \text{ for all } x \in \left[\frac{\pi}{8}, \frac{3\pi}{8} \right]$$

Define $S_i = \int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} f(x) \cdot g_i(x) dx, \quad i = 1, 2$

माना $g_i : \left[\frac{\pi}{8}, \frac{3\pi}{8}\right] \rightarrow R, i = 1, 2$, तथा $f : \left[\frac{\pi}{8}, \frac{3\pi}{8}\right] \rightarrow R$ फलन है जो इस प्रकार हैं कि सभी $x \in \left[\frac{\pi}{8}, \frac{3\pi}{8}\right]$ के लिए

$g_1(x) = 1, g_2(x) = |4x - \pi|$ तथा $f(x) = \sin^2 x$ है। $S_i = \int_{\frac{\pi}{8}}^{\frac{3\pi}{8}} f(x) \cdot g_i(x) dx, \quad i = 1, 2$ को परिभाषित कीजिए।

36. The value of $\frac{16S_1}{\pi}$ is _____. [JEE(Adv.)2021]

$\frac{16S_1}{\pi}$ का मान _____ है। [JEE(Adv.)2021]

Ans. 2

Sol.

37. The value of $\frac{48S_2}{\pi^2}$ is _____. [JEE(Adv.)2021]

$\frac{48S_2}{\pi^2}$ का मान _____ है। [JEE(Adv.)2021]

Ans. 1.5

Sol $S_1 = \int_{\pi/8}^{3\pi/8} f(x) \cdot 1 dx = \int_{\pi/8}^{3\pi/8} \sin^2 x dx \quad \dots(1)$

apply $\int_a^b f(a+b-x) dx = \int_a^b f(x) dx$

$S_1 = \int_{\pi/8}^{3\pi/8} \cos^2 x dx \quad \dots (2)$

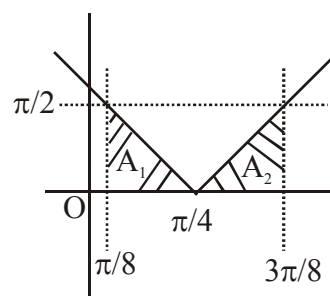
$(1) + (2) \Rightarrow 2S_1 = \int_{\pi/8}^{3\pi/8} dx = \frac{\pi}{4} \Rightarrow S_1 = \frac{\pi}{8}$

$\frac{16S_1}{\pi} = 2$

$$S_2 = \int_{\pi/8}^{3\pi/8} \sin^2 x |4x - \pi| dx \quad \dots\dots(3)$$

$$\text{apply } \int_a^b f(a+b-x) dx = \int_a^b f(x) dx$$

$$S_2 = \int_{\pi/8}^{3\pi/8} \cos^2 x |4x - \pi| dx \quad \dots(4)$$



$$(3) + (4) \Rightarrow 2S_2 = \int_{\pi/8}^{3\pi/8} |4x - \pi| dx = \frac{\pi^2}{16}$$

$$A_1 + A_2 = \frac{1}{2} \times \frac{\pi}{4} \times \frac{\pi}{2} = \frac{\pi^2}{16}$$

$$\Rightarrow S_2 = \frac{\pi^2}{32}$$

$$\Rightarrow \frac{48}{\pi^2} S_2 = 1.5$$

Paragraph for Question Nos. 38 to 39

Let $\psi_1 : [0, \infty) \rightarrow R$, $\psi_2 : [0, \infty) \rightarrow R$, $f : [0, \infty) \rightarrow R$ and $g : [0, \infty) \rightarrow R$ be functions such that $f(0) = g(0) = 0$,

$$\psi_1(x) = e^{-x} + x, \quad x \geq 0,$$

$$\psi_2(x) = x^2 - 2x - 2e^{-x} + 2, \quad x \geq 0,$$

$$f(x) = \int_{-x}^x (|t| - t^2)e^{-t^2} dt, \quad x > 0$$

$$\text{and } g(x) = \int_0^{x^2} \sqrt{t}e^{-t} dt, \quad x > 0.$$

माना $\psi_1 : [0, \infty) \rightarrow R$, $\psi_2 : [0, \infty) \rightarrow R$, $f : [0, \infty) \rightarrow R$ तथा $g : [0, \infty) \rightarrow R$ फलन हैं जो इस प्रकार हैं कि $f(0) = g(0) = 0$,

$$\psi_1(x) = e^{-x} + x, \quad x \geq 0,$$

$$\psi_2(x) = x^2 - 2x - 2e^{-x} + 2, \quad x \geq 0,$$

$$f(x) = \int_{-x}^x (|t| - t^2)e^{-t^2} dt, \quad x > 0$$

$$\text{तथा } g(x) = \int_0^{x^2} \sqrt{t}e^{-t} dt, \quad x > 0.$$

38. Which of the following statements is TRUE?

[JEE(Adv.)2021]

(A) $f(\sqrt{\ln 3}) + g(\sqrt{\ln 3}) = \frac{1}{3}$

(B) For every $x > 1$, there exists an $\alpha \in (1, x)$ such that $\psi_1(x) = 1 + \alpha x$

(C) For every $x > 0$, there exists a $\beta \in (0, x)$ such that $\psi_2(x) = 2x(\psi_1(\beta) - 1)$

(D) f is an increasing function on the interval $\left[0, \frac{3}{2}\right]$

निम्न में से कौनसा कथन सत्य है?

[JEE(Adv.)2021]

(A) $f(\sqrt{\ln 3}) + g(\sqrt{\ln 3}) = \frac{1}{3}$

(B) प्रत्येक $x > 1$ के लिए, एक $\alpha \in (1, x)$ इस प्रकार विद्यमान है कि $\psi_1(x) = 1 + \alpha x$

(C) प्रत्येक $x > 0$ के लिए एक, $\beta \in (0, x)$ इस प्रकार विद्यमान है कि $\psi_2(x) = 2x(\psi_1(\beta) - 1)$

(D) अन्तराल $\left[0, \frac{3}{2}\right]$ में f एक वर्धमान फलन है

Ans. C

Sol. $f(x) = 2 \int_0^x (t - t^2) e^{-t^2} dt$

$$f'(x) = 2(x - x^2)e^{-x^2}$$

$$g'(x) = 2x^2 e^{-x^2}$$

Option (A)

Let consider

$$f'(x) + g'(x) = 2(x - x^2)e^{-x^2} + 2x^2 e^{-x^2}$$

Integrating both side

$$f(x) + g(x) = -e^{-x^2} + c$$

put $x = 0$

then $c = 1$

$$f(x) + g(x) = 1 - e^{-x^2}$$

$$f(\sqrt{\ln 3}) + g(\sqrt{\ln 3}) = 1 - e^{-\ln 3} = 1 - \frac{1}{3} = \frac{2}{3}$$

hence option (A) is wrong.

Option (B)

Let consider

$$H(x) = \psi_1(x) - 1 - \alpha x$$

$$H(1) = \psi_1(1) - 1 - \alpha$$

$$H(1) = e^{-1} + 1 - 1 - \alpha$$

$$H(1) = e^{-1} - \alpha < 0$$

$$H'(x) = \psi_1'(x) - \alpha$$

$$H'(x) = -e^{-x} + 1 - \alpha < 0$$

hence $H(x)$ is decreasing.

So $H(x)$ is always negative and $H(x) \neq 0$

Hence option (B) is wrong.

Option (C)

$$\psi_2(x) = x^2 - 2x - 2e^{-x} + 2$$

Applying LMVT in $(0, x)$

$$\psi_2'(\beta) = \frac{\psi_2(x) - \psi_2(0)}{x - 0}$$

$$\psi_2'(\beta) = \frac{\psi_2(x) - 0}{x} \dots\dots\dots(1)$$

$$\psi_2'(x) = 2x - 2 + 2e^{-x}$$

$$\psi_2'(\beta) = 2\beta - 2 + 2e^{-\beta}$$

$$\psi_2'(\beta) = 2(\beta + e^{-\beta}) - 2$$

$$\psi_2'(\beta) = 2\psi_1(\beta) - 2$$

Now by (1)

$$2(\psi_1(\beta) - 1) = \frac{\psi_2(x)}{x}$$

$$\psi_2(x) = 2x(\psi_1(\beta) - 1)$$

hence option (C) is correct.

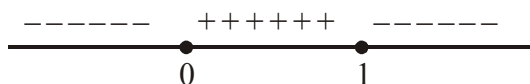
Option (D)

$$f(x) = \int_{-x}^x (|t| - t^2)e^{-t^2} dt$$

$$f(x) = 2 \int_0^x (t - t^2)e^{-t^2} dt$$

$$f'(x) = 2(x - x^2)e^{-x^2}$$

$$f'(x) = 2x(1 - x)e^{-x^2}$$



So option (D) is wrong.

39. Which of the following statements is TRUE ?

[JEE(Adv.)2021]

(A) $\psi_1(x) \leq 1$, for all $x > 0$

(B) $\psi_2(x) \leq 0$, for all $x > 0$

(C) $f(x) \geq 1 - e^{-x^2} - \frac{2}{3}x^3 + \frac{2}{5}x^5$, for all $x \in \left(0, \frac{1}{2}\right)$

(D) $g(x) \leq \frac{2}{3}x^3 - \frac{2}{5}x^5 + \frac{1}{7}x^7$, for all $x \in \left(0, \frac{1}{2}\right)$

निम्न में से कौनसा कथन सत्य है?

[JEE(Adv.)2021]

(A) $\psi_1(x) \leq 1$, सभी $x > 0$ के लिए

(B) $\psi_2(x) \leq 0$, सभी $x > 0$ के लिए

(C) $f(x) \geq 1 - e^{-x^2} - \frac{2}{3}x^3 + \frac{2}{5}x^5$, for all $x \in \left(0, \frac{1}{2}\right)$

(D) $g(x) \leq \frac{2}{3}x^3 - \frac{2}{5}x^5 + \frac{1}{7}x^7$, सभी $x \in \left(0, \frac{1}{2}\right)$ के लिए

Ans. D

Sol. Option (A)

$$\psi_1(x) = e^{-x} + x$$

$$\psi_1'(x) = -e^{-x} + 1$$

$$\psi_1'(x) > 0$$

hence $\psi_1(x)$ is increasing

then

$$\psi_1(x) \geq \psi_1(0)$$

$$\psi_1(x) \geq 1$$

So option (A) is wrong.

Option (B)

$$\psi_2(x) = x^2 - 2x - 2e^{-x} + 2$$

$$\psi_2'(x) = 2x - 2 + 2e^{-x}$$

$$\psi_2'(x) > 0$$

hence $\psi_2'(x)$ is increasing.

then

$$\psi_2(x) \geq \psi_2(0)$$

$$\psi_2(x) \geq 0$$

hence option (B) is wrong.

Option (C)

$$f(x) \geq 1 - e^{-x^2} - \frac{2}{3}x^3 + \frac{2}{5}x^5$$

Consider

$$H(x) = f(x) - 1 + e^{-x^2} + \frac{2}{3}x^3 - \frac{2}{5}x^5$$

$$H'(x) = f'(x) - 2xe^{-x^2} + 2x^2 - 2x^4$$

$$\text{but } f'(x) = 2(x - x^2)e^{-x^2}$$

Now

$$H'(x) = 2x^2(1 - x^2 - e^{-x^2})$$

$$H'(x) = 2x^2(1 - x^2 - 1 + \frac{x^2}{1} - \frac{x^4}{2} + \dots) < 0$$

then $H(x)$ is decreasing

$$H(x) \leq H(0)$$

$$H(x) \leq f(0)$$

$$H(x) \leq 0$$

hence

$$f(x) \leq 1 - e^{-x^2} - \frac{2}{3}x^3 + \frac{2}{5}x^5$$

hence option (C) is wrong.

Option (D)

$$g(x) \leq \frac{2}{3}x^3 - \frac{2}{5}x^5 + \frac{1}{7}x^7$$

Let consider

$$H(x) = g(x) - \frac{2}{3}x^3 + \frac{2}{5}x^5 - \frac{1}{7}x^7$$

$$H'(x) = g'(x) - 2x^2 + 2x^4 - x^6$$

$$\text{but } g'(x) = 2x^2e^{-x^2}$$

$$H'(x) = 2x^2e^{-x^2} - 2x^2 + 2x^4 - x^6$$

$$H'(x) = 2x^2 \left(1 - \frac{x^2}{1} + \frac{x^4}{2} - \frac{x^6}{3} \dots \right) - 2x^2 + 2x^4 - x^6 < 0$$

$H(x)$ is decreasing.

So

$$H(x) \leq H(0)$$

$$g(x) - \frac{2}{3}x^3 + \frac{2}{5}x^5 - \frac{1}{7}x^7 \leq 0$$

$$g(x) \leq \frac{2}{3}x^3 - \frac{2}{5}x^5 + \frac{1}{7}x^7$$

hence option (D) is correct.

40. For any real number x , let $[x]$ denote the largest integer less than or equal to x . If $I = \int_0^{10} \left[\sqrt{\frac{10x}{x+1}} \right] dx$, then the value of $9I$ is _____. [JEE(Adv.)2021]

किसी वास्तविक संख्या x के लिए, माना $[x]$ के बराबर या इससे कम महत्तम पूर्णांक को निरूपित करता है। यदि $I = \int_0^{10} \left[\sqrt{\frac{10x}{x+1}} \right] dx$,

तब $9I$ का मान _____ है। [JEE(Adv.)2021]

Ans. 182

Sol. $f(x) = \frac{10x}{x+1}$

$$f'(x) = \frac{10}{(x+1)^2} > 0$$

hence $f(x)$ is increasing

Now

$$I = \int_0^{\frac{1}{9}} 0 dx + \int_{\frac{1}{9}}^{\frac{2}{3}} 1 dx + \int_{\frac{2}{3}}^9 2 dx + \int_9^{10} 3 dx = \frac{182}{9}$$

$$\text{Now } 9I = 182$$

$$\text{Ans} = 182$$

41. Consider the equation $\int_1^e \frac{(\log_e x)^{1/2}}{x(a - (\log_e x)^{3/2})^2} dx = 1, a \in (-\infty, 0) \cup (1, \infty)$. Which of the following statements is/are

TRUE?

[JEE(Adv.)2022]

- (A) No a satisfies the above equation
(B) An integer a satisfies the above equation
(C) An irrational number a satisfies the above equation
(D) More than one a satisfy the above equation

समीकरण $\int_1^e \frac{(\log_e x)^{1/2}}{x(a - (\log_e x)^{3/2})^2} dx = 1, a \in (-\infty, 0) \cup (1, \infty)$ पर विचार कीजिए। निम्न कथनों में से कौन सा सत्य है?

[JEE(Adv.)2022]

- (A) कोई भी a उपरोक्त समीकरण को संतुष्ट नहीं करता है
(B) एक पूर्णांक (integer) a उपरोक्त समीकरण को संतुष्ट करता है
(C) एक अपरिमेय संख्या (irrational number) a उपरोक्त समीकरण को संतुष्ट करता है
(D) एक से ज्यादा a उपरोक्त समीकरण को संतुष्ट करते हैं

Ans. CD

Sol. Let $I = \int_1^e \frac{(\ln x)^{1/2} dx}{x(a - (\ln x)^{3/2})^2}$

put $a - (\ln x)^{3/2} = t$

$$\Rightarrow -\frac{3}{2}(\ln x)^{1/2} \cdot \frac{1}{x} dx = dt$$

$$\therefore I = \int_a^{a-1} \frac{\left(-\frac{2}{3}\right) dt}{t^2}$$

$$= \left(-\frac{2}{3}\right) \frac{t^{-2+1}}{-2+1} \Big|_a^{a-1}$$

$$= \frac{2}{3t} \Big|_a^{a-1} = \frac{2}{3} \left(\frac{1}{a-1} - \frac{1}{a} \right)$$

$$\therefore I = \left(\frac{2}{3}\right) \frac{1}{a(a-1)} = 1$$

$$\Rightarrow 2 = 3a^2 - 3a$$

$$\Rightarrow 3a^2 - 3a - 2 = 0$$

$$\Rightarrow a = \frac{3 \pm \sqrt{9 - 4(3)(-2)}}{6}$$

$$a = \frac{3 + \sqrt{33}}{6}, \frac{3 - \sqrt{33}}{6}$$

42. The greatest integer less than or equal to $\int_1^2 \log_2(x^3 + 1) dx + \int_1^{\log_2 9} (2^x - 1)^{\frac{1}{3}} dx$ is _____.

[JEE(Adv.)2022]

वह महत्तम पूर्णांक, जो $\int_1^2 \log_2(x^3 + 1) dx + \int_1^{\log_2 9} (2^x - 1)^{\frac{1}{3}} dx$ से कम या बराबर हो _____ है।

[JEE(Adv.)2022]

Ans. 5

Sol. $y = \int_1^2 \log_2(x^3 + 1) dx + \int_1^{2\log_2 3} (2^x - 1)^{\frac{1}{3}} dx$
 $= 4\log_2 3 - 1$

we know $\int_a^b f(x) dx + \int_{f(a)}^{f(b)} (f^{-1}(x)) dx = bf(b) - af(a)$

So, $[y] = 5$

43. Let $f : (0, 1) \rightarrow \mathbb{R}$ be the function defined as $f(x) = \sqrt{n}$ if $x \in \left[\frac{1}{n+1}, \frac{1}{n}\right)$ where $n \in \mathbb{N}$. Let $g : (0, 1) \rightarrow \mathbb{R}$ be

a function such that $\int_{x^2}^x \sqrt{\frac{1-t}{t}} dt < g(x) < 2\sqrt{x}$ for all $x \in (0, 1)$. Then $\lim_{x \rightarrow 0} f(x)g(x) :$ [JEE(Adv.)2023]

(A) does NOT exist

(B) is equal to 1

(C) is equal to 2

(D) is equal to 3

माना कि फलन $f : (0, 1) \rightarrow \mathbb{R}$ इस तरह से परिभाषित है कि $f(x) = \sqrt{n}$ यदि $x \in \left[\frac{1}{n+1}, \frac{1}{n}\right)$ जहाँ $n \in \mathbb{N}$ है। माना कि फलन

$g : (0, 1) \rightarrow \mathbb{R}$ इस प्रकार है कि सभी $x \in (0, 1)$ के लिए $\int_{x^2}^x \sqrt{\frac{1-t}{t}} dt < g(x) < 2\sqrt{x}$ है। तब $\lim_{x \rightarrow 0} f(x)g(x) :$

[JEE(Adv.)2023]

(A) का अस्तित्व नहीं है

(B) 1 के बराबर है

(C) 2 के बराबर है

(D) 3 के बराबर है

Ans. C

Sol. $f(x) = \sqrt{x}$
 $\frac{1}{n+1} \leq x < \frac{1}{n}$
 $n < \frac{1}{x} \leq n+1$

$$n < \frac{1}{x}$$

$$n \geq \frac{1}{x} - 1$$

$$n \geq \frac{1-x}{x}$$

$$\frac{1-x}{x} \leq n < \frac{1}{x}$$

$$\sqrt{\frac{1-x}{x}} \leq \sqrt{n} < \frac{1}{\sqrt{x}}$$

$$\sqrt{\frac{1-x}{x}} \leq f(x) < \frac{1}{\sqrt{x}}$$

$$\int_{x^2}^x \sqrt{\frac{1-t}{t}} dt \times \sqrt{\frac{1-x}{x}} \leq f(x)g(x) < \frac{1}{\sqrt{x}} \times 2\sqrt{x}$$

$$\lim_{x \rightarrow 0} \int_{x^2}^x \sqrt{\frac{1-t}{t}} dt \sqrt{\frac{1-x}{x}} \leq \lim_{x \rightarrow 0} f(x)g(x) < \lim_{x \rightarrow 0} 2$$

$$\lim_{x \rightarrow 0} \frac{\int_{x^2}^x \left(\sqrt{\frac{1-t}{t}} \right) dt}{\sqrt{\frac{x}{1-x}}} \left(\frac{0}{0} \right)$$

L.H. Rule

$$\lim_{x \rightarrow 0} \frac{\sqrt{\frac{1-x}{x}} - \sqrt{\frac{1-x^2}{x^2}} \times (2x)}{\frac{1}{2\sqrt{\frac{x}{1-x}}} \times \left(\frac{1-x+x}{(1-x)^2} \right)} = \lim_{x \rightarrow 0} \frac{2\sqrt{\frac{1-x}{x}} \left(1 - \sqrt{\frac{1+x}{x}} \times 2x \right)}{\sqrt{\frac{1-x}{x}} \times \frac{1}{(1-x)^2}} = 2$$

44. Let $f: [1, \infty) \rightarrow \mathbb{R}$ be a differentiable function such that $f(1) = \frac{1}{3}$ and $3 \int_1^x f(t) dt = xf(x) - \frac{x^3}{3}, x \in [1, \infty)$. Let e denote the base of the natural logarithm. Then the value of $f(e)$ is : **[JEE(Adv.)2023]**

माना कि $f: [1, \infty) \rightarrow \mathbb{R}$ एक अवकलनीय (differentiable) फलन इस प्रकार है कि $f(1) = \frac{1}{3}$ है एवं $x \in [1, \infty)$ के लिए

$3 \int_1^x f(t) dt = xf(x) - \frac{x^3}{3}$ है। माना कि e प्राकृतिक लघुगणक के आधार (base of the natural logarithm) को निरूपित

करता है। तब $f(e)$ का मान है :

[JEE(Adv.)2023]

$$(A) \frac{e^2 + 4}{3}$$

$$(B) \frac{\log_e 4 + e}{3}$$

$$(C) \frac{4e^2}{3}$$

$$(D) \frac{e^2 - 4}{3}$$

Ans. C

Sol. Diff. w.r.t 'x'

$$3f(x) = f(x) + xf'(x) - x^2$$

$$\frac{dy}{dx} - \left(\frac{2}{x}\right)y = x$$

$$IF = e^{-2\ln x} = \frac{1}{x^2}$$

$$y\left(\frac{1}{x^2}\right) = \int x \cdot \frac{1}{x^2} dx$$

$$y = x^2 \ln x + cx^2$$

$$\therefore y(1) = \frac{1}{3} \Rightarrow c = \frac{1}{3}$$

$$y(e) = \frac{4e^2}{3}$$

45. For $x \in \mathbb{R}$, let $\tan^{-1}(x) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then the minimum value of the function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \int_0^{x \tan^{-1} x} \frac{e^{(t - \cos t)}}{1 + t^{2023}} dt \text{ is } \underline{\hspace{2cm}}.$$

[JEE(Adv.)2023]

माना कि $x \in \mathbb{R}$ के लिए $\tan^{-1}(x) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ है। तब फलन $f: \mathbb{R} \rightarrow \mathbb{R}$ जो $f(x) = \int_0^{x \tan^{-1} x} \frac{e^{(t - \cos t)}}{1 + t^{2023}} dt$ से परिभाषित है का

न्यूनतम मान (Minimum value) है $\underline{\hspace{2cm}}$

[JEE(Adv.)2023]

Ans. 0

$$\text{Sol. } f'(x) = \frac{\left(e^{x \tan^{-1} x - \cos(x \tan^{-1} x)}\right)}{1 + (x \tan^{-1} x)^{2023}} \left(\tan^{-1} x + \frac{x}{1 + x^2}\right)$$

$$f'(x) = 0$$

$$\Rightarrow \tan^{-1} x + \frac{x}{1+x^2} = 0 \Rightarrow \boxed{x=0}$$

$$\text{and } f'(0^+) > 0$$

$$f'(0^-) < 0$$

So min at $x=0$

$$\boxed{f(0)=0} \quad \text{Ans.} = 0$$

46. Let the function $f: [1, \infty) \rightarrow \mathbb{R}$ be defined by

$$f(t) = \begin{cases} (-1)^{n+1} 2, & \text{if } t = 2n-1, n \in \mathbb{N} \\ \frac{(2n+1-t)}{2} f(2n-1) + \frac{(t-(2n-1))}{2} f(2n+1), & \text{if } 2n-1 < t < 2n+1, n \in \mathbb{N} \end{cases}$$

Define $g(x) = \int_1^x f(t) dt, x \in (1, \infty)$. Let α denote the number of solutions of the equation $g(x) = 0$ in the interval

$(1, 8]$ and $\beta = \lim_{x \rightarrow 1^+} \frac{g(x)}{x-1}$. Then the value of $\alpha + \beta$ is equal to _____. [JEE(Adv.)2024]

माना कि फलन $f: [1, \infty) \rightarrow \mathbb{R}$

$$f(t) = \begin{cases} (-1)^{n+1} 2, & \text{यदि } t = 2n-1, n \in \mathbb{N} \\ \frac{(2n+1-t)}{2} f(2n-1) + \frac{(t-(2n-1))}{2} f(2n+1), & \text{यदि } 2n-1 < t < 2n+1, n \in \mathbb{N} \end{cases}$$

द्वारा परिभाषित है। $g(x) = \int_1^x f(t) dt, x \in (1, \infty)$ से परिभाषित कीजिए। माना कि α , अंतराल $(1, 8]$ में समीकरण $g(x) = 0$ के

हलों (solutions) की संख्या को दर्शाता है और $\beta = \lim_{x \rightarrow 1^+} \frac{g(x)}{x-1}$ है। तब $\alpha + \beta$ का मान _____ है। [JEE(Adv.)2024]

Ans. 5

Sol. If $t = 2n-1, n \in \mathbb{N}$

$$f(t) = (-1)^{n+1} \cdot 2$$

$$\Rightarrow f(1) = 2, f(3) = -2, f(5) = 2, f(7) = -2, f(9) = 2, \dots$$

$$\text{If } t \in (2n-1, 2n+1)$$

$$f(t) = \frac{(2n+1-t)}{2} f(2n-1) + \frac{(t-(2n-1))}{2} f(2n+1)$$

Now $n = 1$

$$1 < t < 3$$

$$f(t) = \frac{(3-t)}{2}(2) + \frac{(t-1)}{2}(-2)$$

$$\Rightarrow 3 - t - t + 1$$

$$\Rightarrow 4 - 2t$$

now $n = 2$

$$3 < t < 5$$

$$f(t) = \frac{(5-t)}{2}(-2) + \left(\frac{t-3}{2}\right)(2)$$

$$= -5 + t + t - 3$$

$$\Rightarrow -8 + 2t$$

$n = 3$

$$5 < t < 7$$

$$f(t) = \left(\frac{7-t}{2}\right)(2) + \left(\frac{t-5}{2}\right)(-2)$$

$$= 7 - t - t + 5$$

$$\Rightarrow 12t - 2t$$

$n = 4$

$$7 < t < 9$$

$$f(t) = \left(\frac{9-t}{2}\right)(-2) + \left(\frac{t-7}{2}\right)(2)$$

$$= t - 9 + t - 7$$

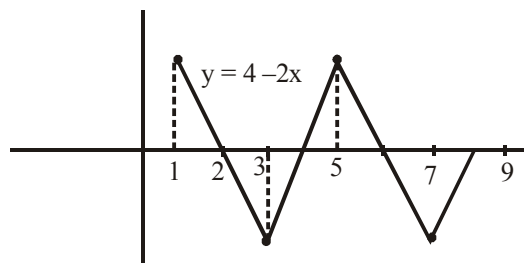
$$\Rightarrow 2t - 16$$

Plot $f(x)$

No. of solution of $g(x) = 3 = \alpha$

$$\beta = \lim_{x \rightarrow 1^+} \frac{g(x)}{x-1}$$

$$= \lim_{x \rightarrow 1^+} g'(x)$$



$$\Rightarrow f(x) \Rightarrow f(1^+) \Rightarrow 2$$

$$\alpha + \beta$$

$$\Rightarrow 3 + 2$$

$$\Rightarrow 5$$

PARAGRAPH

Paragraph for Q. No 47 to 48

Let $f : \left[0, \frac{\pi}{2}\right] \rightarrow [0, 1]$ be the function defined by $f(x) = \sin^2 x$ and let $g : \left[0, \frac{\pi}{2}\right] \rightarrow [0, \infty)$ be the function defined

$$\text{by } g(x) = \sqrt{\frac{\pi x}{2} - x^2}.$$

[JEE(Adv.)2024]

(There are two questions based on PARAGRAPH)

माना कि $f : \left[0, \frac{\pi}{2}\right] \rightarrow [0, 1]$ वह फलन (function) है जो $f(x) = \sin^2 x$ द्वारा परिभाषित है, और माना कि $g : \left[0, \frac{\pi}{2}\right] \rightarrow [0, \infty)$ वह

फलन है जो $g(x) = \sqrt{\frac{\pi x}{2} - x^2}$ द्वारा परिभाषित है।

[JEE(Adv.)2024]

(PARAGRAPH पर आधारित दो प्रश्न हैं, नीचे दिया गया प्रश्न उनमें से एक है)

47. The value of $2 \int_0^{\frac{\pi}{2}} f(x)g(x)dx - \int_0^{\frac{\pi}{2}} g(x)dx$ is _____.

$2 \int_0^{\frac{\pi}{2}} f(x)g(x)dx - \int_0^{\frac{\pi}{2}} g(x)dx$ का मान _____ है।

Ans. 0

Sol. Let $I_1 = 2 \int_0^{\frac{\pi}{2}} \sin^2 x \sqrt{\frac{\pi x}{2} - x^2} dx$ _____ (1)

King

$$I_1 = 2 \int_0^{\frac{\pi}{2}} \cos^2 x \sqrt{\frac{\pi x}{2} - x^2} dx$$
 _____ (2)

(1) + (2)

$$2I_1 = 2 \int_0^{\frac{\pi}{2}} \sqrt{\frac{\pi x}{2} - x^2} dx$$

$$I_1 = \int_0^{\frac{\pi}{2}} \sqrt{\frac{\pi x}{2} - x^2} dx$$

CHALLENGER EXERCISES

Exercise C - I

SINGLE CORRECT TYPE QUESTIONS

1. If $\int \frac{\cos x - \sin x + 1 - x}{e^x + \sin x + x} dx = \ln(f(x)) + g(x) + C$ where C is the constant of integration and $f(x)$ is positive then $f(x) + g(x)$ has the value equal to :

यदि $\int \frac{\cos x - \sin x + 1 - x}{e^x + \sin x + x} dx = \ln(f(x)) + g(x) + C$ जहाँ C समाकलन अचर है तथा $f(x)$ धनात्मक तब $f(x) + g(x)$ का

मान बराबर :

- (A) $e^x + \sin x + 2x$ (B) $e^x + \sin x$ (C) $e^x - \sin x$ (D) $e^x + \sin x + x$

Ans. B

Sol. $I = \int \frac{(e^x + \cos x + 1) - (e^x + \sin x + x)}{e^x + \sin x + x} dx$
 $= \ln(e^x + \sin x + x) - x + C$
 $\therefore f(x) = e^x + \sin x + x$ and $g(x) = -x$
 $f(x) + g(x) = e^x + \sin x$

2. $\int (\sin(101x) \cdot \sin^{99} x) dx$ equals :

$\int (\sin(101x) \cdot \sin^{99} x) dx$ बराबर :

- (A) $\frac{\sin(100x)(\sin x)^{100}}{100} + C$ (B) $\frac{\cos(100x)(\sin x)^{100}}{100} + C$
(C) $\frac{\cos(100x)(\cos x)^{100}}{100} + C$ (D) $\frac{\sin(100x)(\sin x)^{101}}{101} + C$

Ans. A

Sol. $I = \int (\sin(100x + x) \cdot (\sin x)^{99} dx = \int [(\sin(100x) \cos x + \cos 100x \cdot \sin x)] (\sin x)^{99} dx$
 $= \int \underbrace{\sin(100x) \cos x}_I \cdot \underbrace{(\sin x)^{99}}_{II} dx + \int \cos(100x) \cdot (\sin x)^{100} dx$
 $= \frac{\sin(100x)(\sin x)^{100}}{100} - \frac{100}{100} \int \cos(100x)(\sin x)^{100} dx + \int \cos(100x)(\sin x)^{100} dx$
 $= \frac{\sin(100x)(\sin x)^{100}}{100} + C$

3. $\int a^x \left(\ln x + \ln a \cdot \ln \left(\frac{x}{e} \right)^x \right) dx =$

$$(A) a^x \ln\left(\frac{e}{x}\right)^{2x} + C$$

$$(B) a^x \ln\left(\frac{x}{e}\right)^x + C$$

$$(C) a^x + \ln\left(\frac{x}{e}\right)^x + C$$

(D) None of these/इनमें से कोई नहीं

Ans. B

$$\text{Sol. } \int a^x \left(\ln x + \ln a \cdot \ln\left(\frac{x}{e}\right)^x \right) dx$$

$$= \int \frac{d}{dx} \left(a^x \ln\left(\frac{x}{e}\right)^x \right) dx$$

$$a^x \ln\left(\frac{x}{e}\right)^x + C$$

4. $\int \frac{x^2 - 1}{x^3 \sqrt{2x^4 - 2x^2 + 1}} dx$ is equal to :

$\int \frac{x^2 - 1}{x^3 \sqrt{2x^4 - 2x^2 + 1}} dx$ बराबर होगा :

$$(A) \frac{\sqrt{2x^4 - 2x^2 + 1}}{x^2} + C$$

$$(B) \frac{\sqrt{2x^4 - 2x^2 + 1}}{x^3} + C$$

$$(C) \frac{\sqrt{2x^4 - 2x^2 + 1}}{x} + C$$

$$(D) \frac{\sqrt{2x^4 - 2x^2 + 1}}{2x^2} + C$$

Ans. D

$$\text{Sol. } \int \frac{x^2 - 1}{x^3 \sqrt{2x^4 - 2x^2 + 1}} dx = \int \frac{\left(\frac{1}{x^3} - \frac{1}{x^5}\right) dx}{\sqrt{2 - \frac{2}{x^2} + \frac{1}{x^4}}}$$

$$\text{Let } 2 - \frac{2}{x^2} + \frac{1}{x^4} = t \Rightarrow \left(\frac{4}{x^3} - \frac{4}{x^5}\right) dx = dt$$

$$= \frac{1}{4} \int \frac{dt}{\sqrt{t}} = \frac{1}{2} \sqrt{t} + C = \frac{1}{2x^2} \sqrt{2x^4 - 2x^2 + 1} + C$$

5. $\int \frac{8x^{43} + 13x^{38}}{(x^{13} + x^5 + 1)^4} dx =$

(A) $\frac{x^{39}}{3(x^{13} + x^5 + 1)^3} + C$

(B) $\frac{x^{39}}{(x^{13} + x^5 + 1)^3} + C$

(C) $\frac{x^{39}}{5(x^{13} + x^5 + 1)^5} + C$

(D) None of these/इनमें से कोई नहीं

Ans. A

Sol. $\int \frac{8x^{43} + 13x^{38}}{(x^{13} + x^5 + 1)^4} dx$
 $= \int \frac{8x^{-9} + 13x^{-14}}{(1 + x^{-8} + x^{-13})^4} dx$

Let $1 + x^{-8} + x^{-13} = t$

$\Rightarrow -(8x^{-9} + 13x^{-14}) dx = dt$

$= -\int \frac{1}{t^4} dt = \frac{1}{t^3} + C$

$= \frac{1}{3(1 + x^{-8} + x^{-13})^3} + C = \frac{x^{39}}{3(x^{13} + x^5 + 1)^3} + C$

6. $\int_0^1 \frac{\tan^{-1} x}{x} dx =$

(A) $\int_0^{\pi/4} \frac{\sin x}{x} dx$

(B) $\int_0^{\pi/2} \frac{x}{\sin x} dx$

(C) $\frac{1}{2} \int_0^{\pi/2} \frac{x}{\sin x} dx$

(D) $\frac{1}{2} \int_0^{\pi/4} \frac{x}{\sin x} dx$

Ans. C

Sol. $I = \int_0^1 \frac{\tan^{-1} x}{x} dx$ $x = \tan \theta ; dx = \sec^2 \theta d\theta$

$I = \int_0^{\pi/4} \frac{\theta \cdot \sec^2 \theta}{\tan \theta} d\theta = \int_0^{\pi/4} \frac{\theta}{\sin \theta \cos \theta} d\theta = \int_0^{\pi/4} \frac{2\theta}{\sin 2\theta} d\theta$ $2\theta = y$

$$= \frac{1}{2} \int_0^{\pi/2} \frac{y}{\sin y} dy = \frac{1}{2} \int_0^{\pi/2} \frac{x}{\sin x} dx \Rightarrow (C)$$

7. The value of the definite integral $\int_0^{\pi/2} (\sin x + \cos x) \cdot \sqrt{\frac{e^x}{\sin x}} dx$ equals :

निश्चित समाकलन $\int_0^{\pi/2} (\sin x + \cos x) \cdot \sqrt{\frac{e^x}{\sin x}} dx$ का मान है :

(A) $2\sqrt{e^{\pi/2}}$ (B) $\sqrt{e^{\pi/2}}$ (C) $2\sqrt{e^{\pi/2}} \cdot \cos 1$ (D) $\frac{1}{2}e^{\pi/4}$

Ans. A

Sol. $I = \int_0^{\pi/2} \frac{e^x (\sin x + \cos x)}{\sqrt{e^x \sin x}} dx$

put $e^x \sin x = t^2 \Rightarrow e^x (\sin x + \cos x) dx = 2t dt$

$\therefore I = 2 \int_0^{e^{\sqrt{\pi/2}}} \frac{dt}{2t} = 2e^{\sqrt{\pi/2}} = 2\sqrt{e^{\pi/2}}$

8. The absolute value of $\frac{\int_0^{\pi/2} (x \cos x + 1)e^{\sin x} dx}{\int_0^{\pi/2} (x \sin x - 1)e^{\cos x} dx}$ is equal to :

$\frac{\int_0^{\pi/2} (x \cos x + 1)e^{\sin x} dx}{\int_0^{\pi/2} (x \sin x - 1)e^{\cos x} dx}$ का निरपेक्ष मान होगा :

(A) e (B) πe (C) $e/2$ (D) π/e

Ans. A

Sol. $\frac{\int_0^{\pi/2} (x \cos x + 1)e^{\sin x} dx}{\int_0^{\pi/2} (x \sin x - 1)e^{\cos x} dx} = \frac{\left[e^{\sin x} \cdot x \right]_0^{\pi/2}}{\left[-e^{\cos x} \cdot x \right]_0^{\pi/2}}$

$$= \frac{e \cdot \frac{\pi}{2}}{-\frac{\pi}{2}} = -e$$

∴ absolute value is e.

9. If $\int_0^1 \frac{e^t dt}{1+t} = A$ then the value of $\int_0^1 \frac{e^t dt}{(1+t)^2}$ is :

यदि $\int_0^1 \frac{e^t dt}{1+t} = A$ तो $\int_0^1 \frac{e^t dt}{(1+t)^2}$ का मान है :

(A) $A + \frac{e}{2} - 1$ (B) $A - \frac{e}{2} + 1$ (C) $A - \frac{e}{2} - 1$ (D) $A + \frac{e}{2} + 1$

Ans. B

Sol. $I = \int_0^1 \frac{e^t (t+1-t)}{(1+t)^2} dt = \int_0^1 \frac{e^t}{1+t} dt - \int_0^1 e^t \left(\frac{1}{1+t} - \frac{1}{(1+t)^2} \right) dt$
 $= A - \left[\frac{e^t}{1+t} \right]_0^1 = A - \frac{e}{2} + 1$; Alternatively I. B. P. directly

10. If $\beta + 2 \int_0^1 x^2 e^{-x^2} dx = \int_0^1 e^{-x^2} dx$ then the value of β is :

(A) e^{-1} (B) e (C) $1/2e$ (D) can not be determined

यदि $\beta + 2 \int_0^1 x^2 e^{-x^2} dx = \int_0^1 e^{-x^2} dx$ तो β का मान होगा :

(A) e^{-1} (B) e (C) $1/2e$ (D) ज्ञात नहीं किया जा सकता है

Ans. A

Sol. $\beta + \int_0^1 \underbrace{x}_{\text{I}} \underbrace{2xe^{-x^2}}_{\text{II}} dx = \int_0^1 e^{-x^2} dx$

$$\beta + \left[-xe^{-x^2} \right]_0^1 - \int_0^1 -e^{-x^2} dx = \int_0^1 e^{-x^2} dx \quad \beta = \frac{1}{e}$$

11. The value of the definite integral $\int_0^{\pi/2} \sin |2x - \alpha| dx$ where $\alpha \in [0, \pi]$:

निश्चित समाकलन $\int_0^{\pi/2} \sin |2x - \alpha| dx$ का मान ज्ञात कीजिए जहाँ $\alpha \in [0, \pi]$:

- (A) 1 (B) $\cos \alpha$ (C) $\frac{1 + \cos \alpha}{2}$ (D) $(1 + \sin \alpha)$

Ans. A

Sol. $I = \int_{-\alpha}^{(\pi-\alpha)} \sin |t| dt$ where $2x - \alpha = t \Rightarrow dx = \frac{dt}{2}$

$$= \frac{1}{2} \int_{-\alpha}^0 -\sin t dt + \frac{1}{2} \int_0^{\pi-\alpha} \sin t dt = \frac{1}{2} \cos t \Big|_{-\alpha}^0 - \frac{1}{2} \cos t \Big|_0^{\pi-\alpha} = \frac{1}{2} [1 - \cos \alpha] - \frac{1}{2} [-\cos \alpha - 1]$$

$$= \frac{1}{2} (1 - \cos \alpha) + \frac{1}{2} (1 + \cos \alpha) = 1$$

12. The value of the definite integral $\int_2^4 (x(3-x)(4+x)(6-x)(10-x) + \sin x) dx$ equals :

$\int_2^4 (x(3-x)(4+x)(6-x)(10-x) + \sin x) dx$ का मान है :

- (A) $\cos 2 + \cos 4$ (B) $\cos 2 - \cos 4$ (C) $\sin 2 + \sin 4$ (D) $\sin 2 - \sin 4$

Ans. B

Sol. We have $I = \int_2^4 \left(\underbrace{x(3-x)(4+x)(6-x)(10-x)}_E + \sin x \right) dx \dots (1)$

Now $I = \int_2^4 \left((6-x)(3-(6-x))(4+(6-x))(6-(6-x))(10-(6-x)) + \sin(6-x) \right) dx$

(Applying King property)

$$= \int_2^4 \left(\underbrace{(6-x)(x-3)(10-x)x(4+x)}_E + \sin(6-x) \right) dx \dots (2)$$

$\therefore$ On adding (1) and (2), we get $\Rightarrow I = \cos 2 - \cos 4$

13. $\int_{-2}^0 (x^3 + 3x^2 + 3x + 3 + (x+1)\cos(x+1)) dx$ is equal to :

$$\int_{-2}^0 (x^3 + 3x^2 + 3x + 3 + (x+1)\cos(x+1))dx \text{ बराबर होगा :}$$

- (A) -4 (B) 0 (C) 4 (D) 6

Ans. C

Sol. $I = \int_{-2}^0 [x^3 + 3x^2 + 3x + 3 + (x+1)\cos(x+1)]dx$

$$= \left[\frac{x^4}{4} + x^3 + \frac{3x^2}{2} + 3x + (x+1)\sin(x+1) + \cos(x+1) \right]_{-2}^0$$

$$= 4$$

14. Value of the definite integral $\int_{-1/2}^{1/2} (\sin^{-1}(3x - 4x^3) - \cos^{-1}(4x^3 - 3x)) dx :$

$$\int_{-1/2}^{1/2} (\sin^{-1}(3x - 4x^3) - \cos^{-1}(4x^3 - 3x)) dx \text{ का मान होगा :}$$

- (A) 0 (B) $-\frac{\pi}{2}$ (C) $\frac{7\pi}{2}$ (D) $\frac{\pi}{2}$

Ans. B

Sol. $I = \int_{-1/2}^{1/2} (\sin^{-1}(3x - 4x^3) - \cos^{-1}(4x^3 - 3x)) dx$

Apply king property

$$I = \int_{-1/2}^{1/2} (\sin^{-1}(-3x + 4x^3) - \cos^{-1}(-4x^3 + 3x)) dx$$

$$I = \int_{-1/2}^{1/2} -\sin^{-1}(3x - 4x^3) - \pi + \cos^{-1}(4x^3 - 3x) dx$$

$$\Rightarrow 2I = -\pi$$

15. The value of the definite integral $\int_0^{\pi/2} \sqrt{\tan x} dx$, is :

$$\int_0^{\pi/2} \sqrt{\tan x} dx \text{ का मान होगा :}$$

- (A) $\sqrt{2} \pi$ (B) $\frac{\pi}{\sqrt{2}}$ (C) $2\sqrt{2} \pi$ (D) $\frac{\pi}{2\sqrt{2}}$

Ans. B

Sol. $I = \int_0^{\pi/2} \sqrt{\tan x} dx \dots(1); \quad I = \int_0^{\pi/2} \sqrt{\cot x} dx \dots(2)$

adding (1) and (2), we get

$$\begin{aligned} 2I &= \int_0^{\pi/2} (\sqrt{\tan x} + \sqrt{\cot x}) dx = \sqrt{2} \int_0^{\pi/2} \frac{\sin x + \cos x}{\sqrt{\sin 2x}} dx = \sqrt{2} \int_0^{\pi/2} \frac{\sin x + \cos x}{\sqrt{1 - (\sin x - \cos x)^2}} dx \\ &= \sqrt{2} \int_{-1}^1 \frac{dt}{\sqrt{1-t^2}} = 2\sqrt{2} \int_0^1 \frac{dt}{\sqrt{1-t^2}} = \sqrt{2} \pi \quad (\text{where } \sin x - \cos x = t) \\ \therefore I &= \frac{\pi}{\sqrt{2}} \end{aligned}$$

16. If $y(t) = \int_{-lnt}^{lnt} \cos(te^x) dx$, then $y'(1)$ equals :

यदि $y(t) = \int_{-lnt}^{lnt} \cos(te^x) dx$, तो $y'(1)$ का मान है :

- (A) $2 \cos 1$ (B) $-2 \cos 1$ (C) $-\cos 1$ (D) $\cos 1$

Ans. A

Sol. Let $I = \int_{-lnt}^{lnt} \cos(te^x) dx$; Put $te^x = y \Rightarrow t e^x dx = dy$

$$\therefore I = \int_1^{t^2} \frac{\cos(y) dy}{y}$$

$$\therefore y'(t) = \frac{dI}{dt} = \frac{(\cos t^2) 2t}{t^2} - 0 = \frac{2 \cos t^2}{t}$$

So, $y'(1) = 2 \cos 1$

17. $\lim_{x \rightarrow \infty} \left(x^3 \int_{-1/x}^{1/x} \frac{\ln(1+t^2)}{1+e^t} dt \right)$ equals :

$$\lim_{x \rightarrow \infty} \left(x^3 \int_{-1/x}^{1/x} \frac{\ln(1+t^2)}{1+e^t} dt \right) \text{ का मान है :}$$

(A) 1/3

(B) 2/3

(C) 1

(D) 0

Ans. A

Sol. Consider
$$I = \int_{-1/x}^{1/x} \frac{\ln(1+t^2)}{1+e^t} dt \quad \dots(1)$$

$$= \int_{-1/x}^{1/x} \frac{\ln(1+t^2)}{1+e^{-t}} dt \quad (\text{Using King})$$

$$I = \int_{-1/x}^{1/x} \frac{\ln(1+t^2)e^t}{1+e^t} dt \quad \dots(2)$$

(1) + (2)

$$2I = \int_{-1/x}^{1/x} \ln(1+t^2) dt = 2 \int_0^{1/x} \ln(1+t^2) dt \quad \Rightarrow \quad I = \int_0^{1/x} \ln(1+t^2) dt$$

hence
$$I = \lim_{x \rightarrow \infty} x^3 \int_0^{1/x} \ln(1+t^2) dt = \lim_{x \rightarrow \infty} \frac{\int_0^{1/x} \ln(1+t^2) dt}{x^{-3}} \quad \left(\frac{0}{0} \text{ form} \right)$$

Using L'Hospital's Rule

$$I = \lim_{x \rightarrow \infty} \frac{x^4 \ln\left(1 + \frac{1}{x^2}\right) \cdot \left(-\frac{1}{x^2}\right)}{-3} = \frac{1}{3} \lim_{x \rightarrow \infty} x^2 \ln\left(1 + \frac{1}{x^2}\right) = \frac{1}{3} \lim_{x \rightarrow \infty} \ln\left(1 + \frac{1}{x^2}\right)^{x^2} \quad (1^\infty \text{ form})$$

$$= \lim_{x \rightarrow \infty} \frac{1}{3} x^2 \left(1 + \frac{1}{x^2} - 1\right) = \frac{1}{3}$$

18. Let $C_n = \int_{\frac{1}{n+1}}^{\frac{1}{n}} \frac{\tan^{-1}(nx)}{\sin^{-1}(nx)} dx$ then $\lim_{n \rightarrow \infty} n^2 \cdot C_n$ equals :

माना $C_n = \int_{\frac{1}{n+1}}^{\frac{1}{n}} \frac{\tan^{-1}(nx)}{\sin^{-1}(nx)} dx$, तो $\lim_{n \rightarrow \infty} n^2 \cdot C_n$ है :

(A) 1

(B) 0

(C) -1

(D) $\frac{1}{2}$ **Ans.** D

Sol. $C_n = \int_{\frac{1}{n+1}}^{\frac{1}{n}} \frac{\tan^{-1}(nx)}{\sin^{-1}(nx)} dx$ (put $nx = t$) $\Rightarrow C_n = \frac{1}{n} \int_{\frac{1}{n+1}}^{\frac{1}{n}} \frac{\tan^{-1}(t)}{\sin^{-1}(t)} dt$

$$L = \lim_{n \rightarrow \infty} n^2 \cdot C_n = \lim_{n \rightarrow \infty} n \cdot \int_{\frac{1}{n+1}}^{\frac{1}{n}} \frac{\tan^{-1} t}{\sin^{-1} t} dt \quad (\infty \times 0); \quad L = \frac{\int_{\frac{1}{n+1}}^{\frac{1}{n}} \frac{\tan^{-1} t}{\sin^{-1} t} dt}{\frac{1}{n}} \quad \left(\frac{0}{0} \right)$$

applying Leibnitz rule

$$L = \lim_{n \rightarrow \infty} \frac{0 - \frac{\tan^{-1} \frac{n}{n+1}}{\sin^{-1} \frac{n}{n+1}} \left(\frac{1}{(n+1)^2} \right)}{-\frac{1}{n^2}} = \frac{\pi}{4} \cdot \frac{2}{\pi} = \frac{1}{2}$$

19. $\lim_{\lambda \rightarrow 0} \left(\int_0^1 (1+x)^\lambda dx \right)^{1/\lambda}$ is equal to :

$\lim_{\lambda \rightarrow 0} \left(\int_0^1 (1+x)^\lambda dx \right)^{1/\lambda}$ का मान होगा :

(A) $2 \ln 2$ (B) $\frac{4}{e}$ (C) $\ln \frac{4}{e}$

(D) 4

Ans. B

Sol. $\lim_{\lambda \rightarrow 0} \left(\int_0^1 (1+x)^\lambda dx \right)^{1/\lambda} = \lim_{\lambda \rightarrow 0} \left(\frac{(1+x)^{\lambda+1}}{\lambda+1} \Big|_0^1 \right)^{1/\lambda} = \lim_{\lambda \rightarrow 0} \left(\frac{2^{\lambda+1} - 1}{\lambda+1} \right)^{1/\lambda} \quad (1^\infty \text{ form})$

$$= e^{\lim_{\lambda \rightarrow 0} \frac{1}{\lambda} \left(\frac{2^{\lambda+1} - 1 - \lambda - 1}{\lambda+1} \right)} = e^{\lim_{\lambda \rightarrow 0} \left(\frac{2^{\lambda+1} - 2 - \lambda}{\lambda(\lambda+1)} \right)} = e^{\lim_{\lambda \rightarrow 0} \left(\frac{2(2^\lambda - 1)}{\lambda} - 1 \right)} = e^{2 \ln 2 - 1} = e^{\ln \left(\frac{4}{e} \right)} = \frac{4}{e}$$

20. For positive integers $k = 1, 2, 3, \dots, n$, let S_k denotes the area of ΔAOB_k (where 'O' is origin) such that $\angle AOB_k = \frac{k\pi}{2n}$, $OA = 1$ and $OB_k = k$. The value of the $\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=1}^n S_k$ is :

$k = 1, 2, 3, \dots, n$, के लिए माना S_k , ΔAOB_k का क्षेत्रफल (जहाँ 'O' मूल बिन्दु है) इस प्रकार है कि $\angle AOB_k = \frac{k\pi}{2n}$,

$OA = 1$ तथा $OB_k = k$ तो $\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{k=1}^n S_k$ का मान होगा :

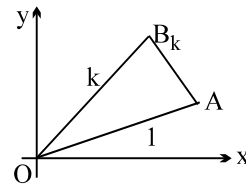
- (A) $\frac{2}{\pi^2}$ (B) $\frac{4}{\pi^2}$ (C) $\frac{8}{\pi^2}$ (D) $\frac{1}{2\pi^2}$

Ans. A

Sol. $OB_k = k$

$$\angle AOB_k = \frac{k\pi}{2n}$$

$$S_k = \frac{1}{2} k \sin \frac{k\pi}{2n} \quad (\text{using } \Delta = \frac{1}{2} ab \sin \theta)$$



$$\therefore L = \frac{k}{2n^2} \sum_{n=1}^{\infty} \sin \frac{k\pi}{2n} = \frac{1}{2n} \sum_{n=1}^{\infty} \frac{k}{n} \sin \frac{k\pi}{2n} = \frac{1}{2} \int_0^1 x \cdot \sin \frac{\pi x}{2} dx$$

$$= \frac{1}{2} \left[\underbrace{\left[\frac{-2}{\pi} x \cos \frac{\pi x}{2} \right]_0^1}_{\text{zero}} + \frac{2}{\pi} \int_0^1 \cos \frac{\pi x}{2} dx \right] = \frac{1}{2} \left[0 + \frac{2}{\pi} \cdot \frac{2}{\pi} \left(\sin \frac{\pi x}{2} \right)_0^1 \right] = \frac{2}{\pi^2}$$

21. The interval $[0, 4]$ is divided into n equal sub-intervals by the points $x_0, x_1, x_2, \dots, x_{n-1}, x_n$ where $0 = x_0 < x_1 < x_2 < x_3 < \dots < x_n = 4$. If $\delta x = x_i - x_{i-1}$ for $i = 1, 2, 3, \dots, n$ then $\lim_{\delta x \rightarrow 0} \sum_{i=1}^n x_i \delta x$ is equal to:

माना अन्तराल $[0, 4]$, n समान उपअन्तरालों में बिन्दुओं $x_0, x_1, x_2, \dots, x_{n-1}, x_n$ द्वारा बांटा जाता है, जहाँ

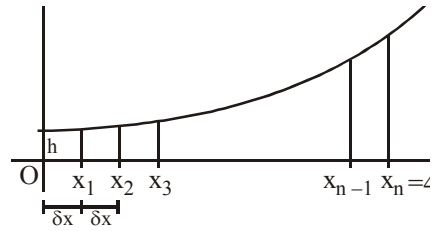
$0 = x_0 < x_1 < x_2 < x_3 < \dots < x_n = 4$. यदि $\delta x = x_i - x_{i-1}$, $i = 1, 2, 3, \dots, n$ के लिए तो $\lim_{\delta x \rightarrow 0} \sum_{i=1}^n x_i \delta x$ का मान होगा :

- (A) 4 (B) 8 (C) $\frac{32}{3}$ (D) 16

Ans. B

Sol. $L = \lim_{\delta x \rightarrow 0} \delta x (x_1 + x_2 + x_3 + \dots + x_n)$

$$= \lim_{n \rightarrow \infty} \frac{4}{n} \left[\frac{4}{n} + \frac{8}{n} + \frac{12}{n} + \dots + 4 \cdot \frac{n}{n} \right] \quad (\delta x = \frac{4}{n})$$



$$= \lim_{n \rightarrow \infty} \frac{16}{n^2} (1 + 2 + 3 + \dots + n) = \lim_{n \rightarrow \infty} \frac{16}{n^2} \cdot \frac{n(n+1)}{2} = 8$$

22. If $f(\theta) = \frac{4}{3}(1 - \cos^6 \theta - \sin^6 \theta)$, then $\lim_{n \rightarrow \infty} \frac{1}{n} \left[\sqrt{f\left(\frac{1}{n}\right)} + \sqrt{f\left(\frac{2}{n}\right)} + \sqrt{f\left(\frac{3}{n}\right)} + \dots + \sqrt{f\left(\frac{n}{n}\right)} \right] =$

यदि $f(\theta) = \frac{4}{3}(1 - \cos^6 \theta - \sin^6 \theta)$ है, तो $\lim_{n \rightarrow \infty} \frac{1}{n} \left[\sqrt{f\left(\frac{1}{n}\right)} + \sqrt{f\left(\frac{2}{n}\right)} + \sqrt{f\left(\frac{3}{n}\right)} + \dots + \sqrt{f\left(\frac{n}{n}\right)} \right] =$

(A) $\frac{1 - \cos 1}{2}$ (B) $1 - \cos 2$ (C) $\frac{\sin 2}{2}$ (D) $\frac{1 - \cos 2}{2}$

Ans. D

Sol. $f(\theta) = \frac{4}{3}(1 - \cos^6 \theta - \sin^6 \theta)$

$$\lim_{n \rightarrow \infty} \frac{1}{n} \left[\sqrt{f\left(\frac{1}{n}\right)} + \sqrt{f\left(\frac{2}{n}\right)} + \sqrt{f\left(\frac{3}{n}\right)} + \dots + \sqrt{f\left(\frac{n}{n}\right)} \right] = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \sqrt{f\left(\frac{r}{n}\right)}$$

$$= \int_0^1 \sqrt{f(x)} dx = \sqrt{\frac{4}{3}} \int_0^1 \sqrt{1 - \cos^6 x - \sin^6 x} dx$$

$$= 2 \int_0^1 \sin x \cos x dx = \left[\frac{\cos 2x}{-2} \right]_0^1$$

$$= \frac{1 - \cos 2}{2}$$

23. Let $f(x)$ be a function satisfying $f'(x) = f(x)$ with $f(0) = 1$ and $g(x)$ be a function that satisfies

$f(x) + g(x) = x^2$. Then the value of the integral $\int_0^1 f(x) g(x) dx$, is :

यदि $f(x)$ फलन इस प्रकार है कि जो $f'(x) = f(x)$ तथा $f(0) = 1$ को संतुष्ट करता है एवं $g(x)$ इस प्रकार है कि जो $f(x) + g(x) =$

x^2 को संतुष्ट करता है तो समाकलन $\int_0^1 f(x) g(x) dx$ का मान होगा :

(A) $e + \frac{e^2}{2} + \frac{5}{2}$ (B) $e - \frac{e^2}{2} - \frac{5}{2}$ (C) $e + \frac{e^2}{2} - \frac{3}{2}$ (D) $e - \frac{e^2}{2} - \frac{3}{2}$

Ans. D

Sol. $f'(x) = f(x)$ and $f(0) = 1 \Rightarrow f(x) = e^x$

$$I = \int_0^1 f(x) g(x) dx = \int_0^1 e^x (x^2 - e^x) dx$$

$$= e - \frac{e^2}{2} - \frac{3}{2}$$

24. The integral $\int \sqrt{\cot x} e^{\sqrt{\sin x}} \sqrt{\cos x} dx$ equals :

$\int \sqrt{\cot x} e^{\sqrt{\sin x}} \sqrt{\cos x} dx$ बराबर :

(A) $\frac{\sqrt{\tan x} e^{\sqrt{\sin x}}}{\sqrt{\cos x}} + C$ (B) $2e^{\sqrt{\sin x}} + C$ (C) $-\frac{1}{2}e^{\sqrt{\sin x}} + C$ (D) $\frac{\sqrt{\cot x} e^{\sqrt{\sin x}}}{2\sqrt{\cos x}} + C$

Ans. B

Sol. $e^{\sqrt{\sin x}} = t$

$$\frac{e^{\sqrt{\sin x}}}{2\sqrt{\sin x}} \cos x ; e^{\sqrt{\sin x}} \sqrt{\cot x} \sqrt{\cos x} dx = 2 dt$$

Hence $\int 2 dt = 2t + C \Rightarrow 2e^{\sqrt{\sin x}} + C$

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

25. $\int_0^\infty \frac{x}{(1+x)(1+x^2)} dx =$

(A) $\frac{\pi}{4}$

(B) $\frac{\pi}{2}$

(C) is same as $\int_0^\infty \frac{dx}{(1+x)(1+x^2)}$

(D) cannot be evaluated

$$\int_0^{\infty} \frac{x}{(1+x)(1+x^2)} dx =$$

(A) $\frac{\pi}{4}$

(B) $\frac{\pi}{2}$

(C) $\int_0^{\infty} \frac{dx}{(1+x)(1+x^2)}$ के समान

(D) ज्ञात नहीं किया जा सकता

Ans. AC

Sol. $I = \int_0^{\infty} \frac{x}{(1+x)(1+x^2)} dx$ _____(1)

let $x = \frac{1}{t}$

$$I = \int_{\infty}^0 \frac{1}{t \left(\frac{t+1}{t} \right) \left(\frac{t^2+1}{t^2} \right)} \times \frac{-1}{t^2} dt$$

$$\Rightarrow I = \int_0^{\infty} \frac{dt}{(t+1)(1+t^2)}$$

$$\Rightarrow I = \int_0^{\infty} \frac{1}{(1+x)(1+x^2)} dx$$
 _____(2)

By (1) + (2)

$$2I = \int_0^{\infty} \frac{dx}{1+x^2} = \left[\tan^{-1} x \right]_0^{\infty} = \frac{\pi}{2}$$

$$\Rightarrow I = \frac{\pi}{4}$$

26. Let $f(x) = \int_0^x e^t \cdot \sin(x-t) dt$ and $g(x) = f(x) + f''(x)$. Which of the following statements are correct?

(A) $g(x)$ is positive $\forall x \in \mathbb{R}$.

(B) $g(x)$ is a constant function

(C) $g(1) = e$

(D) $g'(x) = g(x)$

माना $f(x) = \int_0^x e^t \cdot \sin(x-t) dt$ तथा $g(x) = f(x) + f''(x)$. निम्न में से कौनसे कथन सत्य है?

(A) $g(x)$ घनात्मक है $\forall x \in \mathbb{R}$.

(B) $g(x)$ एक अचर फलन है

(C) $g(1) = e$

(D) $g'(x) = g(x)$

Ans. ACD

Sol. $f(x) = \int_0^x e^t (\sin x \cos t - \cos x \sin t) dt$

$$f(x) = \sin x \int_0^x e^t \cos t dt - \cos x \int_0^x e^t \sin t dt \quad \dots(1)$$

$$f'(x) = \sin x \cdot e^x \cos x + \left(\int_0^x e^t \cos t dt \right) \cos x - \left[\cos x \cdot e^x \sin x - \left(\int_0^x e^t \sin t dt \right) \sin x \right]$$

$$f'(x) = \cos x \int_0^x e^t \cos t dt + \sin x \int_0^x e^t \sin t dt$$

$$f''(x) = \cos x \cdot e^x \cdot \cos x - \sin x \int_0^x e^t \cos t dt + \sin x \cdot e^x \cdot \sin x + \left(\int_0^x e^t \sin t dt \right) \cos x$$

$$f''(x) = e^x + \cos x \int_0^x e^t \sin t dt - \sin x \int_0^x e^t \cos t dt \quad \dots(2)$$

(1) + (2)

$$f(x) + f''(x) = e^x = g(x) \quad \Rightarrow \quad A, C, D$$

27. Let $I_n = \int_0^{\sqrt{3}} \frac{dx}{1+x^n}$ ($n=1, 2, 3, \dots$) and $\lim_{n \rightarrow \infty} I_n = I_0$ (say), then which of the following statement(s) is/are correct?

(Given : $e = 2.71828$)

माना $I_n = \int_0^{\sqrt{3}} \frac{dx}{1+x^n}$ ($n = 1, 2, 3, \dots$) तथा $\lim_{n \rightarrow \infty} I_n = I_0$ (say) तो निम्न में से कौनसे कथन सत्य है?

(दिया गया है : $e = 2.71828$)

(A) $I_1 > I_0$

(B) $I_2 < I_0$

(C) $I_0 + I_1 + I_2 > 3$

(D) $I_0 + I_1 > 2$

Ans. ACD

Sol. We have $I_1 = \ln(1 + \sqrt{3})$; $I_2 = \int_0^{\sqrt{3}} \frac{dx}{1+x^2}$

$$I_2 = \frac{\pi}{3}$$

$$I_0 = \lim_{n \rightarrow \infty} I_n = \lim_{n \rightarrow \infty} \left(\int_0^1 \frac{dx}{1+x^n} + \underbrace{\int_1^{\sqrt{3}} \frac{dx}{1+x^n}}_{\text{zero}} \right) = \int_0^1 dx = 1$$

Hence $I_0 = 1$. Now verify all alternatives

28. Let $I_n = \int_0^{\pi/4} (\tan x)^n dx$ and let $J_n = (-1)^n I_{2n}$ for $n = 0, 1, 2$ then which of the following hold(s) good?

(A) $I_n + I_{n+2} = \frac{1}{n+1}$

(B) $J_n - J_{n-1} = \frac{(-1)^n}{2n-1}$ for $n \geq 1$

(C) If $u = \tan x$ then $I_n = \int_0^1 \frac{u^n}{1+u^2} du$

(D) $\lim_{n \rightarrow \infty} J_n \rightarrow 0$

माना $I_n = \int_0^{\pi/4} (\tan x)^n dx$ and let $J_n = (-1)^n I_{2n}$, $n = 0, 1, 2$ के लिए, तो निम्न में से कौनसे कथन सत्य है?

(A) $I_n + I_{n+2} = \frac{1}{n+1}$

(B) $J_n - J_{n-1} = \frac{(-1)^n}{2n-1}$, $n \geq 1$ के लिये

(C) यदि $u = \tan x$ तो $I_n = \int_0^1 \frac{u^n}{1+u^2} du$

(D) $\lim_{n \rightarrow \infty} J_n \rightarrow 0$

Ans. ABCD

Sol. (A) $I_n + I_{n+2} = \int_0^{\pi/4} \tan^n x \sec^2 x dx = \int_0^1 t^n dt = \frac{1}{n+1} \quad \dots(1) \Rightarrow \quad \text{(A) is correct}$

(B) $J_n - J_{n-1} = (-1)^n I_{2n} - (-1)^{n-1} \cdot I_{2n-2}$
 $= (-1)^n [I_{2n} + I_{2n-2}] = (-1)^n \frac{1}{2n-2+1} \quad (\text{putting } n \rightarrow 2n-2 \text{ in (1)})$
 $= \frac{(-1)^n}{2n-1} \Rightarrow \quad \text{(B) is correct}$

(C) If $u = \tan x \Rightarrow \quad du = \sec^2 x = dx \quad \Rightarrow \quad dx = \frac{du}{1+u^2}$

$\therefore \quad I_n = \int_0^1 \frac{u^n}{1+u^2} du \quad \Rightarrow \quad \text{(C) is also correct}$

(D) obviously correct

29. Which of the following statement(s) is/are TRUE?

(A) If function $y = f(x)$ is continuous at $x = c$ such that $f(c) \neq 0$ then $f(x) f(c) > 0 \forall x \in (c-h, c+h)$ where h is sufficiently small positive quantity.

(B) $\lim_{n \rightarrow \infty} \frac{1}{n} \ln \left(\left(1 + \frac{1}{n}\right) \left(1 + \frac{2}{n}\right) \dots \left(1 + \frac{n}{n}\right) \right) = 1 + 2 \ln 2$.

(C) Let f be a continuous and non-negative function defined on $[a, b]$.

If $\int_a^b f(x) dx = 0$ then $f(x) = 0 \forall x \in [a, b]$.

(D) Let f be a continuous function defined on $[a, b]$ such that $\int_a^b f(x) dx = 0$, then there exists atleast one $c \in (a, b)$ for which $f(c) = 0$

निम्न में से कौनसे कथन सत्य हैं :

(A) यदि फलन $y = f(x)$, $x = c$ पर सतत् है तथा $f(c) \neq 0$ हो तो $f(x) f(c) > 0 \forall x \in (c-h, c+h)$ जहाँ h नगण्य धनात्मक संख्या है।

(B) $\lim_{n \rightarrow \infty} \frac{1}{n} \ln \left(\left(1 + \frac{1}{n}\right) \left(1 + \frac{2}{n}\right) \dots \left(1 + \frac{n}{n}\right) \right) = 1 + 2 \ln 2$.

(C) माना f अन्तराल $[a, b]$ में सतत् तथा अऋणात्मक फलन है। यदि $\int_a^b f(x) dx = 0$ हो तो $f(x) = 0 \forall x \in [a, b]$.

(D) माना फलन f अन्तराल $[a, b]$ में इस प्रकार परिभाषित है कि $\int_a^b f(x) dx = 0$, तो कम से कम एक $c \in (a, b)$ हो जिसके लिए $f(c) = 0$ है।

Ans. ACD

Sol. (A) The expression $f(x) f(c) \forall x \in (c-h, c+h)$ where $h \rightarrow 0^+$ is equivalent to $\lim_{x \rightarrow c} f(x) f(c)$ which equals to $(f(c))^2$ because $f(x)$ is continuous.

$\therefore f(x) f(c) > 0 \forall x \in (c-h, c+h)$ where $h \rightarrow 0^+$.

(B) We have $I = \lim_{n \rightarrow \infty} \frac{1}{n} \ln \left[\left(1 + \frac{1}{n}\right) \left(1 + \frac{2}{n}\right) \dots \left(1 + \frac{n}{n}\right) \right] = \lim_{n \rightarrow \infty} \frac{1}{n} \ln \prod_{k=1}^n \left(1 + \frac{k}{n}\right)$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \ln \left(1 + \frac{k}{n}\right) = \int_1^2 \ln x dx = [x(\ln x - 1)]_{x=1}^{x=2} = -1 + 2 \ln 2 \approx -0.4.$$

(C) Given $f(x) \geq 0 \Rightarrow \int_a^b f(x) dx \geq 0$.

But given $\int_a^b f(x)dx = 0$, so this can be true only when $f(x) = 0$.

$$(D) \int_a^b f(x)dx = 0 \Rightarrow y = f(x) \text{ cuts } x \text{ axis at least once.}$$

So there exists at least one $c \in (a, b)$ for which $f(c) = 0$

30. Let $f(x) = \int_0^x e^{t-[t]} dt$ ($x > 0$), where $[x]$ denotes greatest integer less than or equal to x , is :

- (A) continuous and differentiable $\forall x \in (0, 3]$ (B) continuous but not differentiable $\forall x \in (0, 3]$
 (C) $f(1) = e$ (D) $f(2) = 2(e - 1)$

माना $f(x) = \int_0^x e^{t-[t]} dt$ ($x > 0$), जहाँ $[x]$ महत्तम पूर्णांक फलन है :

- (A) $x \in (0, 3]$ में सतत् तथा अवकलनीय है (B) $x \in (0, 3]$ में सतत् परन्तु अवकलनीय नहीं है
 (C) $f(1) = e$ (D) $f(2) = 2(e - 1)$

Ans. BD

Sol. We have $f(x) = \int_0^x e^{t-[t]} dt = \int_0^x e^{\{t\}} dt$, so

$$f(x) = \begin{cases} \int_0^x e^t dt & \text{if } x \in [0, 1) \\ \int_0^1 e^t dt + \int_1^x e^{t-1} dt & \text{if } x \in [1, 2) \\ \int_0^1 e^t dt + \int_1^2 e^{t-1} dt + \int_2^x e^{t-2} dt & \text{if } x \in [2, 3) \end{cases} \Rightarrow f(x) = \begin{cases} e^x - 1 & \text{if } x \in [0, 1) \\ (e - 1) + (e^{x-1} - 1) & \text{if } x \in [1, 2) \\ 2(e - 1) + (e^{x-2} - 1) & \text{if } x \in [2, 3) \end{cases}$$

Clearly $f(x)$ is continuous $\forall x > 0$ but not differentiable $\forall x \in \mathbb{N} \Rightarrow$ (B)

Also $f(2) = 2(e - 1) = 0 = 2(e - 1) \Rightarrow$ (D)

31. If $a, b, c \in \mathbb{R}^+$ then $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n}{(k+an)(k+bn)}$ is equal to :

- (A) $\frac{1}{a-b} \ln \frac{b(b+1)}{a(a+1)}$ if $a \neq b$ (B) $\frac{1}{a-b} \ln \frac{a(b+1)}{b(a+1)}$ if $a \neq b$
 (C) non existent if $a = b$ (D) $\frac{1}{a(1+a)}$ if $a = b$

यदि $a, b, c \in \mathbb{R}^+$ हो, तो $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n}{(k+an)(k+bn)}$ का मान है :

(A) $\frac{1}{a-b} \ln \frac{b(b+1)}{a(a+1)}$ यदि $a \neq b$

(B) $\frac{1}{a-b} \ln \frac{a(b+1)}{b(a+1)}$ यदि $a \neq b$

(C) यदि $a = b$ तो अस्तित्व में नहीं

(D) $\frac{1}{a(1+a)}$ यदि $a = b$

Ans. BD

Sol. We have $\lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{n}{n^2 \left(a + \frac{k}{n}\right) \left(b + \frac{k}{n}\right)} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^n \frac{1}{\left(a + \frac{k}{n}\right) \left(b + \frac{k}{n}\right)}$

$$= \int_0^1 \frac{1}{(a+x)(b+x)} dx = \frac{1}{a-b} \int_0^1 \frac{(a+x) - (b+x)}{(a+x)(b+x)} dx = \frac{1}{a-b} \int_0^1 \left(\frac{1}{b+x} - \frac{1}{a+x} \right) dx$$

$$= \frac{1}{a-b} [\ln(b+x) - \ln(a+x)]_0^1 = \frac{1}{a-b} \left[\ln \frac{(b+1)a}{(a+1)b} \right]_0^1 = \frac{1}{a-b} \ln \frac{(b+1)a}{(a+1)b} \Rightarrow (B)$$

Now if $a = b$ then $I = \lim_{n \rightarrow \infty} \sum_{k=1}^n \frac{1}{n \left(a + \frac{k}{n}\right)^2} = \int_0^1 \frac{dx}{(a+x)^2} = \left[\frac{1}{a+x} \right]_0^1 = \frac{1}{a} - \frac{1}{a+1} = \frac{1}{a(a+1)} \Rightarrow (D)$

32. If $\int_{-\alpha}^{\alpha} \left(e^x + \cos x \ln \left(x + \sqrt{1+x^2} \right) \right) dx > \frac{3}{2}$, then possible value of α can be :

यदि $\int_{-\alpha}^{\alpha} \left(e^x + \cos x \ln \left(x + \sqrt{1+x^2} \right) \right) dx > \frac{3}{2}$ है, तो α के संभावित मान होंगे :

(A) 1

(B) 2

(C) 3

(D) 4

Ans. ABCD

Sol. $I = \int_{-\alpha}^{\alpha} e^x + \cos x \ln \left(x + \sqrt{1+x^2} \right) dx$

Apply king prop.

$$I = \int_{-\alpha}^{\alpha} \left(e^{-x} - \cos x \ln \left(\sqrt{x^2+1} + x \right) \right) dx$$

$$\Rightarrow 2I = \int_{-\alpha}^{\alpha} \left(e^x + e^{-x} \right) dx$$

$$= 2 \int_0^{\alpha} (e^x + e^{-x}) dx$$

$$\Rightarrow I = [e^{\alpha} - 1] - [e^{-\alpha} - 1]$$

$$\Rightarrow I = \frac{e^{2\alpha} - 1}{e^{\alpha}}$$

$$\text{Given } I > \frac{3}{2}$$

$$\Rightarrow 2e^{2\alpha} - 2 > 3e^{\alpha}$$

$$\text{let } e^{\alpha} = t$$

$$2t^2 - 3t - 2 > 0$$

$$\Rightarrow 2t^2 - 3t - 2 > 0$$

$$\Rightarrow e^{\alpha} < -\frac{1}{2} \text{ or } e^{\alpha} > 2$$

$$\Rightarrow \alpha \text{ can be 1, 2, 3 or 4}$$